The Alan Turing Institute

Centre for Emerging Technology and Security

RESEARCH REPORT

Semiconductor Supply Chains, AI and Economic Statecraft

A framework for UK-Korea strategic cooperation

Ardi Janjeva, Seungjoo Lee, Harish Bhaskaran, Seoin Baek and Hyunjin Lee

April 2024



About CETaS **2**
Acknowledgements **2**
Executive Summary **3**
Summary Recommendations **6**
Glossary **9**
Introduction **11**
1. The Global Context: The Rise of 'Siliconpolitik' **15**
1.1 The trillion-dollar industry on which our societies depend 15
1.2 Export controls: Foregrounding national security 17
1.3 The revival of industrial policy 19
1.4 The impossibility of self-sufficiency 25
2. The Technological and Strategic Relationship between AI and Semiconductors **29**
2.1 The software-hardware relationship 29
2.2 AI challenging the laws of compute 29
2.3 Strategic AI-semiconductor partnerships and collaborations 34
2.4 Looking ahead: A 'frontier AI' dominated landscape? 37
3. The UK Semiconductor Industry: Punching Above its Weight? **39**
3.1 Recent developments 39
3.2 Areas of strength 42
3.3 Hardware and cybersecurity 45
3.4 Leveraging UK R&D 49
4. The South Korea Semiconductor Industry: A National Treasure **53**
4.1 Recent developments 53
4.2 Areas of strength 58
4.3 SK Hynix/Samsung strategic pivots 58
4.4 Leveraging semiconductor strengths for a globally competitive AI industry 62
5. Semiconductors and Geostrategic Alliances: Opportunities for the UK and Korea **64**
5.1 A high-point in UK-Korea relations 64
5.2 Complementarities in semiconductors and AI 66
5.3 Focusing efforts 68
Extended Recommendations **70**
About the Authors **76**

About CETaS

The Centre for Emerging Technology and Security (CETaS) is a research centre based at The Alan Turing Institute, the UK's national institute for data science and artificial intelligence. The Centre's mission is to inform UK security policy through evidence-based, interdisciplinary research on emerging technology issues. Connect with CETaS at cetas.turing.ac.uk.

This research was supported by The Alan Turing Institute's Defence and Security Programme. All views expressed in this report are those of the authors and do not necessarily represent the views of The Alan Turing Institute or any other organisation.

Acknowledgements

The authors wish to thank all those who took part in a research interview or focus group for this project. They are particularly grateful to Professor John Goodenough, Professor Zahid Durrani, and Pranay Kotasthane for their valuable feedback on earlier versions of this report. The quality of the final output is due to their detailed and candid responses.

A special acknowledgement is extended to the Korean co-authors of this report and their associated networks and colleagues. Prior to the commencement of this project, members of the CETaS team visited Korea on two separate occasions, holding meetings with government, industry, and academic stakeholders, and presenting at technology and security conferences. The hospitality extended to the CETaS team on each occasion surpassed all expectations. We are very grateful to our Korean friends, and we hope this project will be the first of many in a long-lasting and fruitful collaboration.

This work is licensed under the terms of the Creative Commons Attribution License 4.0 which permits unrestricted use, provided the original authors and source are credited. The license is available at: https://creativecommons.org/licenses/by-nc-sa/4.0/legalcode.

Cite this work as: Ardi Janjeva, Seungjoo Lee, Harish Bhaskaran, Seoin Baek and Hyunjin Lee, "Semiconductor Supply Chains, AI and Economic Statecraft: A framework for UK-Korea strategic cooperation," *CETaS Research Reports* (April 2024).

Executive Summary

This CETaS Research Report examines the intersection between semiconductors, AI, and global economic security, to provide a framework for strategic cooperation. The study is a collaboration between The Alan Turing Institute and Chung-Ang University, Seoul, with a specific focus on strengthening UK-Korea bilateral partnership.

The disruption inflicted on global semiconductor supply chains by the COVID-19 pandemic emphasised crucial dependencies on sufficient and timely access to semiconductor capability. Semiconductors are now one of five 'critical technologies' for the UK Government, but their position at the bottom of the technology stack means they are the building blocks underpinning progress across almost every other emerging technology. AI is dominating global technology discourse, but it cannot be treated in isolation in a world where the raw materials underpinning its progress are concentrated in individual chip companies widely dispersed across the globe.

These concerns have instigated a wave of policy responses; both in the form of domestic industrial policy and export controls. The UK has mostly watched these developments from the side-lines but is cognisant of the need to partner with like-minded countries that *do* occupy a more dominant position in the supply chain. The UK's 2023 Integrated Review Refresh committed to "work with international partners to diversify supply and make the global semiconductors market more resilient to shock."

Action is needed to bring this to fruition. The current structure is too diffuse, and international partners wishing to work with the UK on semiconductors lack a clear point of engagement. We propose that a new UK National Semiconductor Institute should be established to fulfil this function and avoid confusion as to the UK's strategic ambitions in this sector.

Korea is one of the key countries with which the UK must explore a more ambitious bilateral approach, building on what is already a golden period in UK-Korea relations, following the November 2023 UK-Korea Downing Street Accord. From Korea's perspective, the UK represents a like-minded partner vis-à-vis de-risking from China, while possessing a strong academic research base and cybersecurity expertise, both essential to a competitive chip industry.

Overall, there are significant complementarities between the UK and Korea: the UK has a strategic position in core intellectual property (IP) and chip design, while Korea has significant capabilities in the subsequent manufacturing stage of the supply chain.

Moreover, both countries are showing a desire to be at the vanguard of AI innovation and international discussions concerning AI safety. The second Global AI Safety Summit planned to be co-hosted by Korea and the UK this year is testament to this.

Yet without semiconductor chips, there is no AI. Semiconductors enable the processing of large amounts of data and the rapid execution of complex calculations that form the foundation of AI systems. This is increasingly highlighted with the advent of 'frontier AI' systems requiring advanced AI chips, and recent compute investments and government strategies in both the UK and Korea have reflected this. More broadly, strategic partnerships between AI and semiconductor companies are proliferating at speed, in recognition that the laws of compute that have held for decades are breaking down in an AI-driven technology landscape. The AI and semiconductor industries are increasingly shaping each other's trajectories; UK-Korea collaboration within these industries must cross into the other for the two countries to remain ahead of the curve.

Against this backdrop, the UK must develop a clearer outline of its offering to like-minded international partners. In lieu of wide-ranging industrial policy, it must double down on segments of critical technology supply chains where it holds a unique advantage, such as fabless chip design, and play to these strengths with partners who can compensate for areas where the UK is weaker.

As well as core IP and chip design, the UK's compound semiconductor sector is showing promise, and a wealth of cybersecurity and hardware expertise is giving it a leading role in the 'security for AI' ecosystem. UK academic expertise in next-generation compute and new materials is another vital card to be played – leading on the technologies of 10-20 years' time is a proactive strategic choice, when the winners in the current silicon-dominated paradigm are already clear.

Korean semiconductor giants like Samsung and SK Hynix have been quick to realise the shift in the centre of gravity caused by the AI sector. The domestic investments and partnerships that have been built to secure their position can play a key role in shaping a joint UK-Korea vision for the future of AI.

Korea has a rich history in the semiconductor industry and must now decide how to leverage this to be a global leader in AI. Comparatively, the UK has a rich history of AI start-ups, innovation, and deep tech research, and must now service this for greater semiconductor supply chain resilience. Both countries are pushing at an open door in their quest to deepen ties and strengthen technological competitiveness.

Following in-depth consultation with government, industry, and academic experts across both countries, we propose 4 broad categories within which the UK and Korea should pursue closer partnership: coordinating and convening stakeholders; joint R&D and investment; skills cultivation and training; and international diplomacy. The UK faces tough global competition positioning itself as a partner of choice for the Korean semiconductor industry, but embracing these recommendations would immediately put the UK in a more favourable position, while boosting Korea's supply chain de-risking efforts with a reliable, like-minded Western partner possessing unique strengths of its own.

Summary Recommendations

This is a shortened summary of 15 recommendations – the more detailed descriptions can be found at the end of the report.

Category of partnership	Recommendation
Coordinating and convening stakeholders	1. *Establishing and coordinating national authorities:* **A UK National Semiconductor Institute should be established this year.** Working with the UK Semiconductor Advisory Panel, this body should develop a bespoke engagement strategy with Korea's Advanced Semiconductor Technology Centre. 2. *Coordinating non-government expertise:* The UK Department for Science Innovation (DSIT) and the Korean Technology and Ministry of Science and ICT (MSIT) should sponsor **quarterly meetings between semiconductor industry stakeholders in both countries, expanding this to academic and research institutes as necessary**. 3. *Coordinating AI and Semiconductors portfolios:* The UK and Korean Governments should **develop joint working groups for departments overseeing AI and semiconductor policy respectively (DSIT and the Department for Business and Trade [DBT] in the UK, and MSIT and the Ministry of Trade, Industry and Energy [MOTIE] in Korea).** This could mirror the new '2+2' structure which will bring together the Foreign and Defence Ministries of both countries on an annual basis. 4. *AI-enabled early warning systems:* The Korean Ministry of Foreign Affairs (MOFA) and UK DBT should **develop more formal, real-time information sharing mechanisms relating to supply chain risks and vulnerabilities**, cross-referencing with the OECD's Semiconductor Informal Exchange Network.
Joint R&D and investment	5. *Government funding:* **DSIT and MSIT should ringfence proportions of their international funding for collaborations between UK and Korean research institutes or companies**. Initial funding should prioritise longer-term projects to target the development of mid-career researcher relationships. 6. *Collaboration on new AI compute architectures:* The UK and Korean governments, working closely with industry, should **promote strategic**

	partnerships between companies best positioned to develop GPU-alternative future compute infrastructures. 7. *Advanced packaging facilities:* **A UK National Packaging Institute** should be established, which could be bid for by existing industrial, governmental and university organisations in the UK. The new Institute should **explore the suitability of Korean investment into a new 2.5D/3D hybrid advanced packaging centre in the UK,** while partnering with Korean companies on wafer scale integration. 8. *Reserve capacity:* UK and Korean national semiconductor bodies should liaise with industry to determine whether a **percentage of Korean fabrication capacity can be dedicated to servicing UK start-ups**; in return UK industry should scope which areas of its design capacity could be marketed to Korean companies on a short timescale. 9. *Compound semiconductors:* The UK should pursue plans to create an **open-access compound semiconductor foundry** and **assess the feasibility of Korean investors making up any funding shortfall**. The UK compound semiconductor industry should leverage its leading research base to attract investment as Korean industry leaders seek to build out their compound semiconductor lines. 10. *Cybersecurity/hardware expertise:* As part of the new UK-Korea Strategic Cyber Partnership, the UK Ministry of Defence and Korean defence and security community should **scope hardware security or capability domains that could be suitable for joint R&D or sharing of best practice**, such as radio frequency and advanced mixed signal technology.
Skills cultivation and training	11. *Design IP/tooling capability:* The UK Government should fund **a new Design Competence Centre, with Korea as a priority partner upon international scaling.** This could launch a **UK-Korea moonshot project to create AI-based open-source electronic design automation (EDA) tools** for all aspects of chip design. 12. *Skills:* UK and Korean industry leaders should conduct a **joint mapping exercise identifying skills shortages in their respective semiconductor sectors and devise a two-way upskilling programme** which pools resources into areas with the greatest need.
International diplomacy	13. *Third countries and multilateral fora:* Leveraging the new annual 2+2 Foreign and Defence Ministerial meetings, the UK and Korea should strive for **closer policy alignment in third countries with a strategic position in the semiconductor supply chain,** and each consider how

	they can **advocate for the expansion of certain multilateral fora or agreements to include the other.** 14. *International standards:* **UK-Korea cybersecurity cooperation should be reflected in their positioning on international standards.** The existing R&D partnership between DSIT in the UK and the Institute of Information & Communications Technology Planning and Evaluation (IITP) in Korea designed to accelerate the deployment of Open RAN technologies should be scaled up, taking advantage of Korean industry's central positioning. 15. *AI safety:* **When Korea launches its AI Safety Research Institute, the UK's AISI should be its leading international partner**, aligning closely with the US AISI based in NIST and partners in Europe to **share best practice and develop common approaches.**

Glossary

Semiconductors (or 'chips'): materials with electrical conductivity between that of a conductor and an insulator. Semiconductors form the basis of devices like transistors and integrated circuits and are commonly made from silicon.

Fabrication: the process of manufacturing semiconductor devices and integrated circuits. This involves a series of intricate steps carried out in specialised facilities known as plants or 'fabs'.

Silicon wafers: thin, circular slices of crystalline silicon used as the substrate for manufacturing semiconductor devices.

Transistor: a fundamental semiconductor device used to switch electric signals and electrical power.

Integrated circuit: a miniaturised electronic circuit consisting of interconnected semiconductor devices, such as transistors and resistors, fabricated on a single substrate.

Logic chip: electronic component designed to process data. Examples include microprocessors and digital signal processors.

Memory chip: an integrated circuit that stores and retrieves digital information.

NAND (Negative-AND): a type of non-volatile memory technology used in flash memory. NAND flash memory is widely used for data storage in devices like USB drives.

DRAM (Dynamic Random Access Memory): a type of volatile memory used in computers and other digital devices. It stores data in capacitors within a memory cell, which must be periodically refreshed to maintain the data.

Moore's Law: a prediction made by Gordon Moore, co-founder of Intel Corporation, in 1965 that the number of transistors on a microchip double approximately every two years, leading to a doubling of computational power and performance while reducing the cost per transistor.

Graphics Processing Units (GPUs): specialised electronic circuits designed to accelerate the rendering of images for display on a computer screen. GPUs are now widely used in applications like AI due to their ability to perform parallel processing tasks efficiently.

Central Processing Units (CPUs): the primary components of a computer responsible for executing instructions from software programmes. CPUs typically contain multiple processing cores and cache memory to enhance performance.

Instruction Set Architecture (ISA): the set of instructions that a CPU understands and can execute. Different ISAs dictate CPUs' capabilities and compatibility with software.

Electronic Design Automation (EDA): a category of software tools used in the design and development of electronic systems like integrated circuits and printed circuit boards, streamlining the design process and improving productivity.

Extreme Ultraviolet Lithography (EUV): an advanced semiconductor manufacturing technique used to create extremely small features on silicon wafers during the fabrication of integrated circuits.

Chiplets: individual semiconductor dies that are manufactured separately and then integrated into a single package to form an integrated circuit. These allow semiconductor companies to pair different technologies and manufacturing processes for optimal performance.

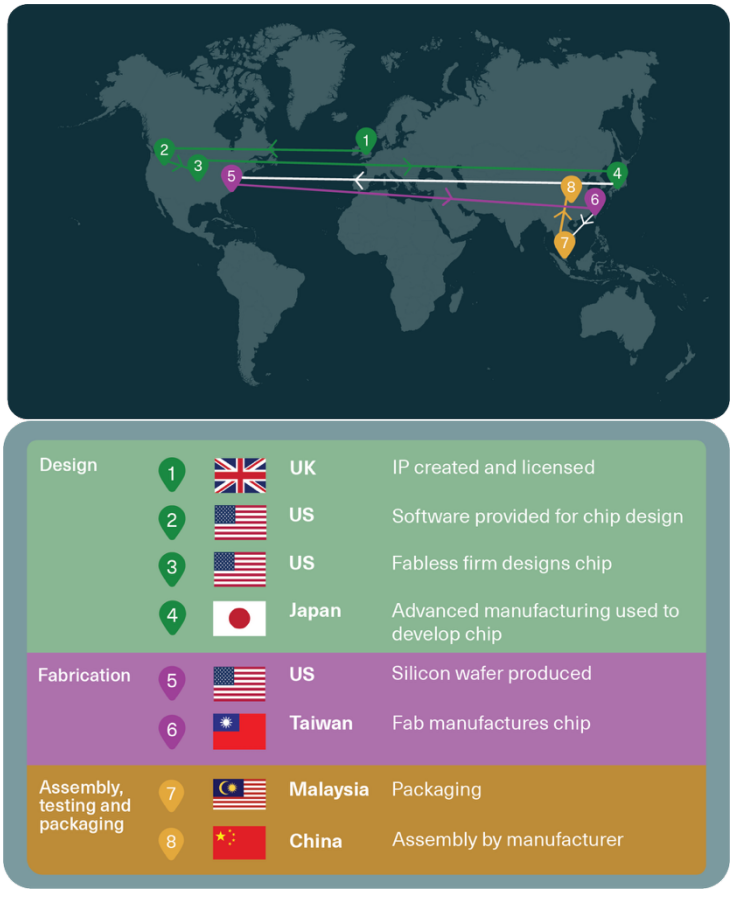
Heterogenous integration: combining different semiconductor materials, devices or technologies into a single integrated circuit or package. This allows for the integration of diverse functionalities, like logic, memory, and sensors into single systems.

Introduction

The semiconductor supply chain

There are more than 30 types of semiconductor product categories, and developing those products requires deep hardware and software expertise, and advanced design tools and IP. Fabrication then involves as many as 300 inputs, processed by more than 50 classes of precision equipment.[1] The result is a complex and highly internationalised supply chain: from chip design to fabrication, and assembly, testing, and packaging (ATP).

Figure 1. The illustrative journey of the smartphone



Source: Adapted from Imagination Technologies and Global Counsel, 2022.[2]

[1] Antonio Varas et al., *Strengthening the Global Semiconductor Supply Chain in an Uncertain Era* (Boston Consulting Group and Semiconductor Industry Association: 2021), 4, https://www.bcg.com/publications/2021/strengthening-the-global-semiconductor-supply-chain.

[2] Imagination Technologies and Global Counsel, The Future of the UK's Semiconductor Strategy (2022), 6, https://6008785.fs1.hubspotusercontent-na1.net/hubfs/6008785/GC_Imagination_18_new%20colour.pdf.

Design

The design phase involves several crucial steps, including specification, logic design, physical design, and validation and verification. In this phase, the functionality and layout of the chip are determined, ensuring it meets the requirements of the system in which it will be used. The US leads in logic chip design, but the UK and India also have a developed talent pool here (the top 10 fabless[3] chipmakers all have R&D centres in India). Key areas of focus in logic chip design include high-end central processing unit (CPU) design, discrete graphics processing unit (GPU) design, field programmable gate arrays (FPGA) design, and AI application-specific integrated circuit (ASIC) design.

Fabrication

Fabrication, or the manufacturing of chips, occurs in semiconductor fabrication facilities (fabs). These produce chips by forming transistors and other electrical devices within silicon wafers, followed by the formation of metal interconnects between these devices. There are two main business models for fabs: those owned by integrated device manufacturers (IDMs) which operate in different areas of the supply chain, and foundries that focus on manufacturing chips for third-party customers. Most fab market share and capacity is controlled by firms headquartered in the US, Taiwan, Korea, Japan, and China. Notably, Intel (US), TSMC (Taiwan), and Samsung (Korea) dominate the world's advanced logic fab capacity, with Taiwan and Korea housing the most cutting-edge fabs. However, the most advanced machines for lithography are primarily manufactured and supplied by ASML, a Dutch company, which supplies over 90% of all such tools.[4]

ATP

Following fabrication, the finished wafer contains multiple chips, which undergo assembly, testing, and packaging. During ATP, the wafer is separated into individual chips, mounted on frames, wired to external devices, and enclosed in protective casings. This stage ensures that each chip operates as intended. As transistor densities in logic chips have increased, packaging has become increasingly critical for chip performance. ATP services are conducted either in-house by IDMs and foundries or by outsourced semiconductor assembly and test (OSAT) firms. Taiwan, the US, China, and Korea are the main providers of ATP services.

---

[3] Fabless chipmakers design and market semiconductors for smartphones and other electronic devices, but they outsource their manufacturing to other parties, which are in turn called foundries.

[4] Toby Sterling, "Focus: Computer Chip Giant ASML Places Big Bets on a Tiny Future," Reuters, 20 May 2022, https://www.reuters.com/technology/computer-chip-giant-asml-places-big-bets-tiny-future-2022-05-20/.

In summary, semiconductor supply chains involve a series of crucial, interconnected stages. Understanding the roles of key players and the challenges within these supply chains is vital for navigating the complexities of the semiconductor industry.

Methodology

This study aimed to address the following research questions:

- **RQ1:** What are the key features of the relationship between the global AI and semiconductor ecosystems, and what are the economic security implications?
- **RQ2:** What is the role of national security communities in using their software and hardware expertise in pursuit of a more resilient semiconductor sector?
- **RQ3:** What would an 'inclusive' approach to technology sovereignty look like in the UK-South Korea bilateral context, and what are the relative advantages possessed by each country that could be leveraged for mutual gain?
- **RQ4:** What long-term bilateral and multi-national policy initiatives are required to ensure long-term economic security in relation to future critical technology supply chains?

To answer these, the research team conducted 23 semi-structured interviews in the UK and 11 in Korea between January and February 2024, including participants across the UK and Korean governments, industry, and academia. These participants were identified through a purposive sampling strategy, and a snowball sampling method identified a limited number of additional interview candidates. A semi-structured interview approach kept the broad line of questioning consistent across interviews while allowing interviewers to elicit further detail from interviewees' specific areas of expertise. Interviews were conducted on an anonymised basis.

The research team conducted a targeted literature review, both in English and Korean, to map key developments in:

- The global semiconductor landscape, including supply chain interdependencies, export controls and industrial policy.
- The relationship between AI and semiconductors over the last 10 years.
- The UK and Korean semiconductor and AI landscapes, including areas of possible competitive advantage.
- Strategies for international cooperation in critical technology areas.

Country visits were also made by the research team, courtesy of a two-part Wilton Park series titled *From Intent to Implementation – Towards comprehensive South Korea-UK economic and technology security and international cooperation*. This ran across 2022-23,

bringing together 40 academics, practitioners, and experts to exchange views on closer collaboration, including in areas like AI and semiconductors. Insights gained from these sessions helped in shaping the authors' thinking and making connections across stakeholder communities.

One limitation of this project was the inability for the research teams in the UK and Korea respectively to be present in each other's research interviews. This allowed interviews to be conducted in first languages and for interviewees to express themselves accordingly, but potentially resulted in slight variations in interview style.

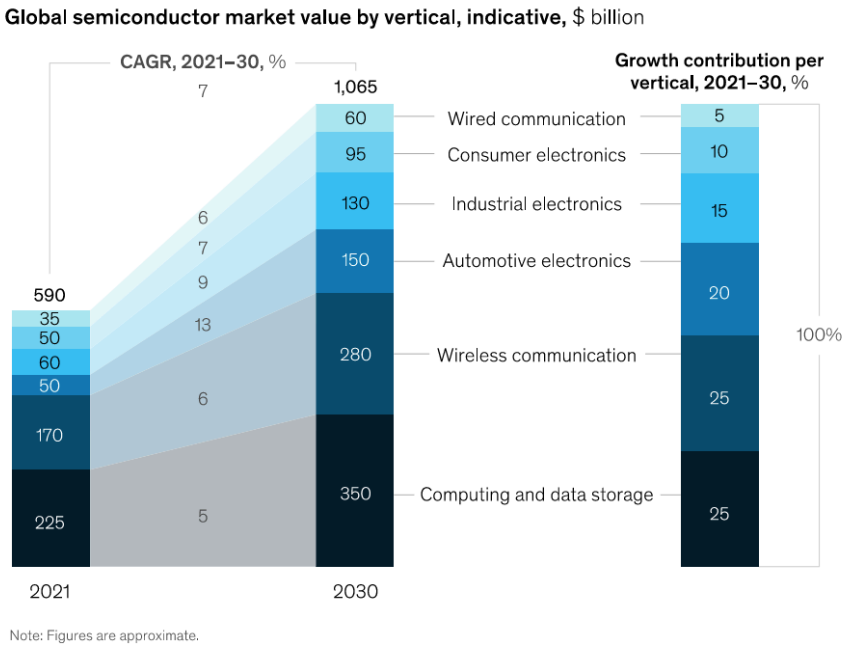
Research participants took part in this study in a personal capacity. The views and responses expressed here reflect participants' opinions and should not be interpreted to represent the official position of any government department, agency, or other organisation.

1. The Global Context: The Rise of 'Siliconpolitik'

1.1 The trillion-dollar industry on which our societies depend

The semiconductor industry is projected to be worth $1 trillion by 2030. From semiconductors' initial development in the US in the 1950s to today, the growth and innovation undergone by the industry has been unprecedented. The number of transistors per wafer for a logic chip has increased by a factor of around 10 million, yielding a 100,000-fold gain in processor speed.[5] The growth of the industry has been defined by 'Moore's Law' – a prediction made by Gordon Moore in 1965 (revised in 1975) that the number of transistors that could fit on a single chip would double roughly every two years.

Figure 2. Semiconductor market projections to 2030 (CAGR = Compound Annual Growth Rate)



Source: McKinsey & Company, 2022.[6]

[5] Antonio Varas et al., Strengthening the Global Semiconductor Supply Chain in an Uncertain Era (Boston Consulting Group and Semiconductor Industry Association: 2021), 5.

[6] Ondrey Burkacky, Julia Dragon and Nikolaus Lehmann "The Semiconductor Decade: A Trillion-Dollar Industry," McKinsey & Company, 1 April 2022, https://www.mckinsey.com/industries/semiconductors/our-insights/the-semiconductor-decade-a-trillion-dollar-industry.

This constant improvement is what has allowed semiconductors to dictate the pace of change in virtually every other technology sector, while bringing into its orbit industries like the automotive industry which previously had little need for chips. For example, autonomous vehicle (AV) driving requires huge volumes of advanced chips in cutting edge data centres to interpret data from sensors and make real-time decisions.[7]

The upshot of semiconductors' ubiquity is that the industry is now too important to fail. Over the next 10 years, industry will need to invest around $3 trillion in R&D across the global value chain to meet increasing demand.[8] Smarter transportation, better medical devices, agricultural efficiencies, climate solutions, and defence capabilities will all depend on better semiconductor technology.[9] For example, while the HIMARS missiles used in Ukraine do not require the most advanced chips themselves, they do rely on targeting information provided by a vast array of sensors and processors to sort the signal from the noise.[10] More broadly, defence and security communities depend on a range of microelectronics, capable of operation in difficult environments involving heat, vibration and radiation challenges.[11]

Global security is now so intertwined with the ability of this industry to maintain its output that disruption in just a handful of fabs around the world can send inflation soaring, while the prospect of vulnerabilities being inserted into semiconductor hardware by malicious actors could ostensibly upend critical national infrastructure (CNI) around the world. The COVID pandemic exposed existing bottlenecks: the industry has evolved in a way which trades off resilience for cost-effectiveness, but this is now increasingly viewed as a strategic handicap by governments.[12] This has led to a wider proliferation of export controls targeting the development of adversaries' military capabilities, and a revival of industrial policy designed to onshore manufacturing capacity.

---

[7] Chris Miller, "Interview: Chris Miller, Historian and Author of "Chip War"," interview by Noah Smith, Noahopinion, 11 July 2023, https://www.noahpinion.blog/p/interview-chris-miller-historian.

[8] Antonio Varas et al., Strengthening the Global Semiconductor Supply Chain in an Uncertain Era (Boston Consulting Group and Semiconductor Industry Association: 2021), 5.

[9] MITRE Engenuity and The Semiconductor Alliance, American Innovation, American Growth: A Vision and Strategy for the NSTC (November 2021), https://f.hubspotusercontent20.net/hubfs/7754670/Semiconductors/A%20Vision%20for%20the%20National%20Semiconductor%20Technology%20Center_MITRE%20Engenuity.pdf.

[10] Chris Miller, "Interview: Chris Miller, Historian and Author of "Chip War"," interview by Noah Smith, Noahopinion, 11 July 2023.

[11] MITRE Engenuity and The Semiconductor Alliance, American Innovation, American Growth: A Vision and Strategy for the NSTC (November 2021), 29; Zahid Durrani and Mervyn Jones, Response to Inquiry on the Semiconductor Industry in the UK (Imperial College London: 2022), 4, https://committees.parliament.uk/writtenevidence/109129/pdf/.

[12] Pranay Kotasthane, "Siliconpolitik: The Case for a Quad Semiconductor Partnership," National University of Singapore, 26 April 2021, https://www.isas.nus.edu.sg/papers/siliconpolitik-the-case-for-a-quad-semiconductor-partnership/.

1.2 Export controls: Foregrounding national security

For the US, export controls represent perhaps the only opportunity to influence Chinese military and intelligence capabilities over the current decade, by hobbling China's ability to catch up in certain parts of the chip industry in the short-to-medium term.[13]

One crucial tool has been the Foreign Direct Product Rule (FDPR). This is a regulation the US initially used to prevent Huawei from rolling out 5G outside China. By May 2020, US firms could no longer provide equipment that would be used to supply Huawei.[14] This applied to offshore, foreign-owned factories too, and made it illegal to use Electronic Design Automation (EDA) software to design chips for Huawei without a license. This soon meant that Huawei lost access to advanced foundries like TSMC, and due to a lack of access to chips, saw falling sales outside China.[15]

In October 2022, new regulations extended these rules beyond Huawei, prohibiting companies from using US code, equipment, or people to make advanced computer chips bound for China.[16] This effectively created jurisdiction over almost every chip factory in the world, marking a shift towards weaponizing the ubiquity of US technology rather than just the dollar. Earlier this year, further restrictions were announced on some US investment into China's quantum computing, advanced chips, and AI sectors, while requiring companies to notify the US Government of other investments in those three sectors – described by US National Security Advisor Jake Sullivan as a 'small yard, high fence' strategy.[17]

The effectiveness of these measures is debated. By using export controls to restrict China's access to leading-edge chips (5 nanometres (nm) and smaller) needed for advanced AI capability, China will be driven to ramp up production volumes for lagging-edge chips (for example around the 28nm node).[18] This is a strategy that carries a distinct risk if the AI hype does not hold.[19] Moreover, at this larger size, export controls have a more limited impact because of the larger number of countries that can produce the necessary tools for production.

---

[13] Chris Miller, "Interview: Chris Miller, Historian and Author of "Chip War"," interview by Noah Smith, Noahopinion, 11 July 2023.
[14] Douglas Fuller, "Navigating the Chip War," Cutter, 11 July 2023, https://www.cutter.com/article/navigating-chip-war.
[15] Douglas Fuller, "Navigating the Chip War," Cutter, 11 July 2023.
[16] The Economist, "The History and Limits of America's Favourite New Economic Weapon," 8 February 2023, https://www.economist.com/united-states/2023/02/08/the-history-and-limits-of-americas-favourite-new-economic-weapon.
[17] Demetri Sevastopulo, "White House Unveils Ban on US Investment in Chinese Tech Sectors," Financial Times, 10 August 2023, https://www.ft.com/content/64ef2042-9ece-4b0c-ad02-184c3454f43b.
[18] Author interview with academic participant, 10 January 2023.
[19] The Economist, "The History and Limits of America's Favourite New Economic Weapon," 8 February 2023.

If the US yardstick of success is degradation of Chinese military capabilities, this will only be measurable over a 5-10-year time horizon.[20] There is no single type of chip without which China's military modernisation suddenly stops, but the more a country needs to design around sub-par semiconductors, the more trade-offs need to be made between key characteristics like performance and reliability.[21] On the other hand, if the measure of success is imposing costs on China's path to technological advancement, then the review is more positive, despite some problems in the enforcement and monitoring of the controls.[22] China is still unable to produce the most advanced equipment – and without links to critical industry players beyond China, the money being pumped into the domestic equipment industry is unlikely to lead to significant catch-up.[23]

Unsurprisingly, China has ways of responding to these measures, given its dominant position across a range of rare earth materials. In response to US measures in 2023, China imposed curbs on exports of gallium and germanium, metals crucial to the chipmaking process, and classified by the US Government as critical to economic and national security.[24] This also resulted in emergency meetings being called by the Korean government, and the EU calling on its aluminium and zinc companies to explore ways of producing those materials.[25]

[20] Author interview with academic participant, 10 January 2023.
[21] Chris Miller, "Interview: Chris Miller, Historian and Author of "Chip War"," interview by Noah Smith, Noahopinion, 11 July 2023.
[22] Author interview with academic participants, 11 January 2023.
[23] Douglas Fuller, "Export Controls Effectively Constrain China's Advance in Microchip Production," Danish Institute for International Studies Policy Brief (August 2023), https://www.diis.dk/en/research/export-controls-effectively-constrain-chinas-advance-in-microchip-production.
[24] Financial Times, "China's Curb on Metal Exports Reverberates Across Chip Sector," 4 July 2023, https://www.ft.com/content/2fa865a7-176f-4292-8842-38bb6470d732.
[25] Harry Dempsey and Alice Hancock, "EU Asks Metals Producers to Explore Making Chip Inputs After China Export Curbs," Financial Times, 14 July 2023, https://www.ft.com/content/b08c11fe-8b09-4819-b8be-02e63b0bf2df.

Figure 3. Rare earth market reserves and production, 2021



Source: Statista, 2023.[26]

From the UK's perspective, being caught up in this type of retaliation while having relatively fewer cards to play represents a risk in its own right. This potentially explains why the tone of the UK Government's National Semiconductor Strategy published in May 2023 was more actor-agnostic than other policy announcements from the likes of the US, Japan, and the Netherlands.[27]

1.3 The revival of industrial policy

The export control regimes described above have operated in parallel with the revival of semiconductor industrial policy, closely tied to the race for AI chips. For decades, US public investment in R&D has remained quite flat, while industry R&D has ballooned. This imbalance has shifted research priorities from long-term breakthroughs towards short-term

[26] Martin Armstrong, "China Dominates the Rare Earth Market," Statista, 13 January 2023, https://www.statista.com/chart/29114/rare-earth-reo-reserves-and-production/.
[27] Author interview with government participants, 22 January 2023.

incremental improvements.[28] The surge in funds for re-shoring parts of the semiconductor supply chain is partly a recognition of the need to reprise government's role in advancing basic research. It is also a recognition of the sheer expense of continual innovation in the semiconductor industry (the cost of advanced chip design almost doubles with each new generation[29]), and the need for the private sector to be incentivised to operate in countries for geopolitical rather than pure economic reasons.

To this effect, Jake Sullivan said the following in a speech in 2022:

> *'We have to revisit the longstanding premise of maintaining 'relative' advantages over competitors in certain key technologies... given the foundational nature of certain technologies, such as advanced logic and memory chips we must maintain as large of a lead as possible.'* [30]

The table below summarises key semiconductor industrial policies enacted in recent years by the US, China, EU, Korea, and India – outlining their associated funding, institutional vehicles, goals, impact, and limitations.

[28] Brady Helwig et al., National Action Plan for United States Leadership in Advanced Compute & Microelectronics (Special Competitive Studies Project: November 2023), 23, https://www.scsp.ai/wp-content/uploads/2023/11/National-Action-Plan-for-U.S.-Advantage-in-Advanced-Compute-and-Microelectronics.pdf.

[29] Jamie Gaida et al., ASPI's Critical Technology Tracker: The Global Race for Future Power (Australian Strategic Policy Institute: 2023), 23, https://ad-aspi.s3.ap-southeast-2.amazonaws.com/2023-08/ASPIs%20Critical%20Technology%20Tracker.pdf.

[30] Jake Sullivan, "Remarks by National Security Advisor Jake Sullivan at the Special Competitive Studies Project Global Emerging Technologies Summit," speech by Jake Sullivan, The White House, 16 September 2022, https://www.whitehouse.gov/briefing-room/speeches-remarks/2022/09/16/remarks-by-national-security-advisor-jake-sullivan-at-the-special-competitive-studies-project-global-emerging-technologies-summit/.

Table 1. Summary of global semiconductor industrial policies

Government policy	Funding vehicles	Goal	Impact	Limitations
 **United States: *CHIPS* and *Science Act***	Split $39bn for manufacturing incentives across the microelectronics value chain and $11bn for microelectronics R&D.[31] Plus, a refundable 25% advanced manufacturing investment tax credit until 2026. Funding established a new National Semiconductor Technology Center and a National Advanced Packaging Manufacturing Programme.[32]	If planned investments materialise, the US will make 26% of the world's advanced chips by 2027, up from 10% now.[33]	TSMC announced $40bn investment in a new Arizona fab, with production expected to begin this year. According to one estimate, almost 500 companies have applied for US government funding.[34]	Differences in the operating environment between the US and Asia mean that TSMC manufacturing investments in Japan are coming online quicker than in the US.[35] Plus, advanced chip production in Taiwan will remain one generation ahead of US-based production.

[31] The White House, "FACT SHEET: CHIPS and Science Act Will Lower Costs, Create Jobs, Strengthen Supply Chains, and Counter China," 9 August 2022, https://www.whitehouse.gov/briefing-room/statements-releases/2022/08/09/fact-sheet-chips-and-science-act-will-lower-costs-create-jobs-strengthen-supply-chains-and-counter-china/.

[32] The White House, "FACT SHEET: Biden-Harris Administration Announces Over $5 Billion from the CHIPS and Science Act for Research, Development, and Workforce," 9 February 2024, https://www.whitehouse.gov/briefing-room/statements-releases/2024/02/09/fact-sheet-biden-harris-administration-announces-over-5-billion-from-the-chips-and-science-act-for-research-development-and-workforce/.

[33] Lauly Li and Cheng Ting-Fang, "How Taiwan Became the Indispensable Economy," Nikkei Asia, 31 May 2023, https://asia.nikkei.com/static/vdata/infographics/taiwan-economy/.

[34] Hannah Mullane, "Why is the World Investing So Much in Semiconductors?," BBC News, 12 January 2024, https://www.bbc.co.uk/news/business-67576918.

[35] Arthur Sants, "What TSMC's New Plant Tells Us About Intel and the US's Ambitions," Investors' Chronicle, 5 March 2024, https://www.investorschronicle.co.uk/news/2024/03/05/what-tsmc-s-new-plant-tells-us-about-intel-and-the-us-s-ambitions/.

 **China: *Made in China 2025* (various)**	Although obtaining conclusive figures is challenging, and *Made in China 2025* is focused on a wide range of sectors, one report estimated the Chinese government invested $290.8bn into semiconductor-related sectors in 2021-22, with one-third going to semiconductor equipment and materials.[36] A wide range of Ministries and local governments are mobilised to administer funding and implement policies.	Increase the Chinese-domestic content of core materials to 70% by 2025. By 2049 the goal is to be a leading world manufacturing power.[37]	China's chip production capacity may grow 60% in the next 3 years via a massive buildout for lagging-edge chip nodes,[38] potentially increasing Western countries' dependence on China in sectors which rely on legacy chips (28nm-180nm range).[39]	China still needs willing buyers in the West, and the EU, Japan and US appear increasingly reluctant markets for China's electronic vehicle (EV) exports.[40] At the same time, if the CCP pressures domestic firms to buy lower quality Chinese chips instead of foreign ones, there would likely be big repercussions for global trade.
 **European Union: *European Chips Act***	The Act mobilises more than €43bn of public and private investment. Pillars 1, 2 and 3 of the Act respectively are being implemented by the following: The Chips Joint	To double the EU's global market share in semiconductors from 10% to 20% by 2030[42]; production on European soil of	Chips JU funding activities start in 2024 for several pilot lines aimed at producing new chips.[44] One notable investment is Intel's €30bn	Initial targets are unlikely to be met. For example, TSMC will not produce its leading-edge chips in Germany. Also, there is a risk of subsidies disproportionately favouring Germany, contradicting the spirit of European competition law.[46]

[36] Qianer Liu, "How Huawei Surprised the US with a Cutting-edge Chip Made in China," Financial Times, 30 November 2023, https://www.ft.com/content/327414d2-fe13-438e-9767-333cdb94c7e1.
[37] Paul McNamara, "Made in China 2025: Separating Fact From Fiction," Supplyframe, https://intelligence.supplyframe.com/made-in-china-2025-fact-fiction/.
[38] Chris Miller, "Western Nations Need a Plan for when China Floods the Chip Market," Financial Times, 28 January 2024, https://www.ft.com/content/2bd1c1a3-931a-4e95-9ea2-e1e8c635ff50.
[39] Brady Helwig et al., National Action Plan for United States Leadership in Advanced Compute & Microelectronics (Special Competitive Studies Project: November 2023), 9.
[40] Douglas Fuller, "Tech War or Phony War? China's Response to America's Controls on Semiconductor Fabrication Equipment," China Leadership Monitor, Issue 78 (December 2023): 8.
[42] Kathrin Hille and Laura Pitel, "TSMC to Build €10bn Chip Plant in Germany," Financial Times, 8 August 2023, https://www.ft.com/content/ee01aae1-06ee-43f6-ad14-caacdffab93e.
[44] Jari Kinaret, "The Smallest Objects of Utmost Importance to the EU," interview by EUobserver, 21 November 2023, https://euobserver.com/stakeholders/157713.
[46] Mathieu Duchatel, "TSMC in Germany: The Successes and Limits of the EU Chips Act," Euractiv, 10 August 2023.

	Undertaking (JU) via Horizon Europe, the Framework for Integrated Production Facilities and Open EU Foundries, and the European Semiconductor Board.[41]	2nm generation chips.[43]	investment in Magdeburg, Germany, the largest single foreign direct investment in European history.[45]	
 **South Korea: *K-Chips Act*; *National Advanced Strategic Industry Act***	*K-Chips Act* raised corporate tax break for facility investment in the semiconductor industry from 8% to 15% for large companies, and 25% from 16% for SMEs.[47] K-Chips legislation was passed and later revised in the South Korea National Assembly, via the *Restriction of Special Taxation Act*.	South Korea's top 10 chipmakers may save up to $277 million for every percentage point increase in the tax deduction rate.[48]	Samsung announced a $230bn investment in a massive chip cluster in Yongin, Gyeonggi Province, by 2042. The site will become the world's largest semiconductor manufacturing site by production capacity.[49]	Tax credit incentives do not directly address the estimated 30,000 worker shortage for the Korean semiconductor industry over the next 10 years.[50]
	India has dedicated $10bn in incentives, the majority of which is to encourage local chip	To build a semiconductor manufacturing base from scratch and become the	The Tata Group and Taiwanese Foundry PSMC are partnering to establish a fab in Gujarat which will	Some other planned investments have stalled or been scrapped, and the incentives do not necessarily outweigh concerns around broader trade policy which is

[41] "The European Chips Act," last accessed 3 April 2024, https://www.european-chips-act.com.

[43] Mathieu Duchatel, "TSMC in Germany: The Successes and Limits of the EU Chips Act," Euractiv, 10 August 2023, https://www.euractiv.com/section/industrial-strategy/opinion/tsmc-in-germany-the-successes-and-limits-of-the-eu-chips-act/.

[45] Nick Alipour, "Intel Stands by Record Semiconductor Investment Despite Germany Budget Problems," Euractiv, 5 December 2023, https://www.euractiv.com/section/politics/news/intel-stands-by-record-semiconductor-investment-despite-german-budget-problems/.

[47] Jo He-Rim, "Korean Chips Act Aims to Extend Tax Cuts for Local Chipmakers," The Korea Herald, 30 March 2023, https://www.koreaherald.com/view.php?ud=20230330000782.

[48] Jo He-Rim, "Korean Chips Act Aims to Extend Tax Cuts for Local Chipmakers," The Korea Herald, 30 March 2023.

[49] Park Eun-Jee, "Samsung Electronics to Invest $230 Billion Into Chip Complex," Korea JoongAng Daily, 15 March 2023, https://koreajoongangdaily.joins.com/2023/03/15/business/tech/Korea-Samsung-Electronics-chip-cluster/20230315180555286.html.

[50] Alex Kim, "Chip on the Shoulder: The Impact of the US CHIPS Act on South Korea," Wilson Center Blog, 10 October 2023, https://www.wilsoncenter.org/blog-post/chip-shoulder.

India: *Semicon India* (various)	manufacturing.[51] The India Semiconductor Mission is implementing the $10bn programme.	most viable friend-shoring destination for companies de-risking from China. The component of the funding focused on design-linked incentives aim to fulfil the goal of 'Made in India' chips with Indian IP.	produce 50,000 wafers per month, estimated to cost $11bn. Two other assembly and testing investments in Assam and Gujarat have been approved.[52]	seen as overly protectionist: high import tariffs are a major reason why TSMC have not shown interest in assembly in India.[53]

The UK is notable by its absence in the list above. The 2023 Harrington Review of Foreign Direct Investment concluded that:

> 'Our competitors chase investments via their industrial strategies backed by substantial government support. They identify which 'races' they want to be in, which sectors and sub-sectors they have a competitive advantage in, and how they are going to attract the finest businesses in the world to their country. The UK needs to respond.'[54]

Section 4 will return to the UK's approach in more detail, but what binds the five countries above is the focus on *new investment in manufacturing.* The lack of this in the UK means that developing effective global partnerships carries additional importance in building resilience across the UK semiconductor supply chain.

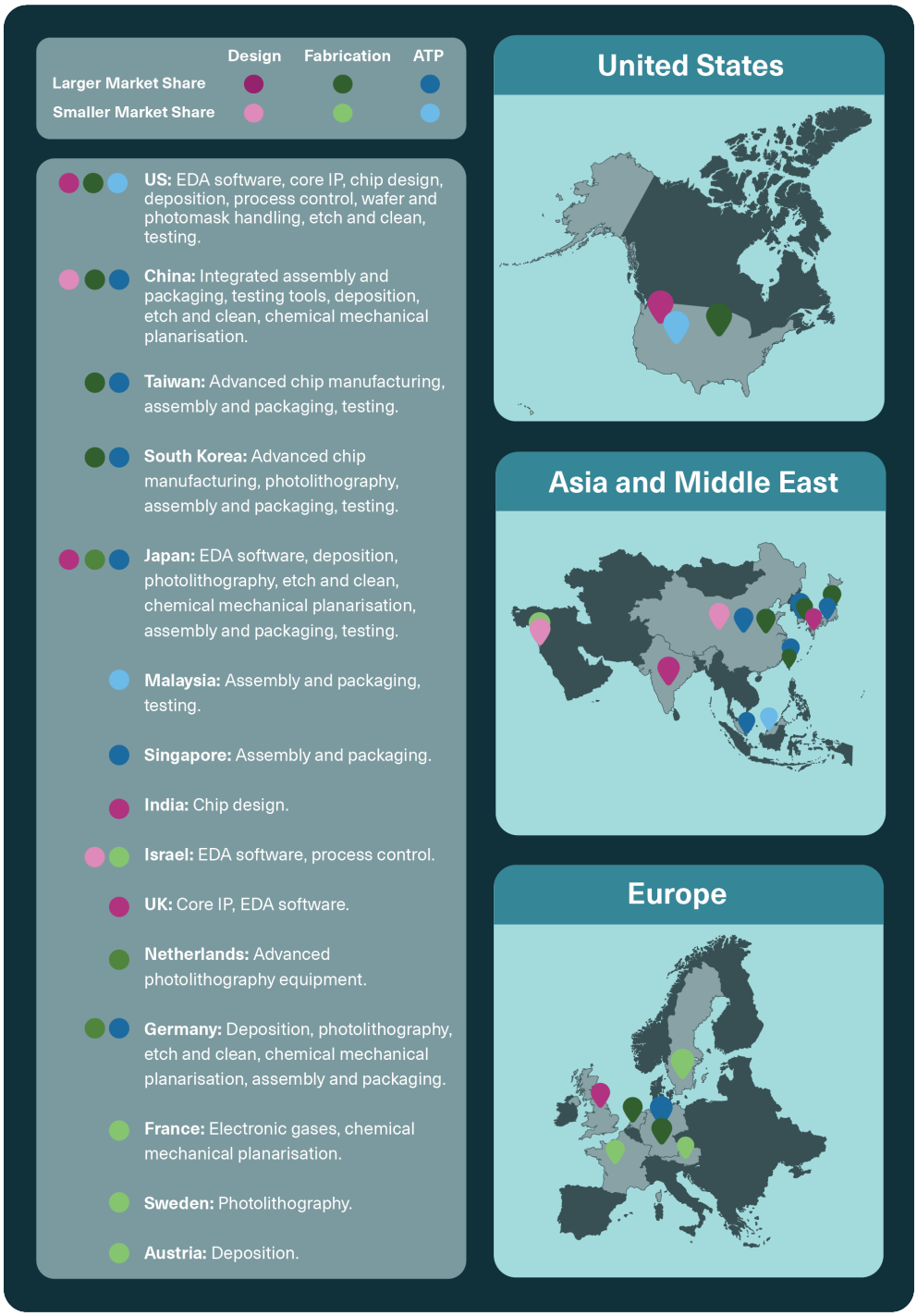
[51] The Economic Times, "Semiconductor PLI Worth $10bn to Help India do in a Decade what China did in 25-30 Years: Rajeev Chandrasekhar," 27 July 2023, https://economictimes.indiatimes.com/industry/cons-products/electronics/semiconductor-pli-worth-10-bn-to-help-india-do-in-a-decade-what-china-did-in-25-30-years-rajeev-chandrasekhar/articleshow/102171938.cms.
[52] Government of India, Giant Leap for India Semiconductor Mission: Cabinet Approves Three More Semiconductor Units (India Ministry of Electronics & IT: February 2024), https://pib.gov.in/PressReleaseIframePage.aspx?PRID=2010135.
[53] Pranay Kotasthane, "$10 Billion Bet: Can India Move Beyond Subsidies in Global Chip Races," interview by Abhiram Manchi, Outlook India, 8 December 2023, https://business.outlookindia.com/economy-and-policy/10-billion-bet-can-india-move-beyond-subsidies-in-global-chip-game.
[54] HM Government, Harrington Review of Foreign Direct Investment (Department for Business and Trade: November 2023), 6, https://assets.publishing.service.gov.uk/media/655f62310c7ec8001195bd5f/231123_Harrington-Review-Report-FINAL-2__HH_Global_.pdf.

1.4 The impossibility of self-sufficiency: global partnerships and the UK's Indo-Pacific tilt

Self-sufficiency in the semiconductor industry is unattainable, and perhaps except for China, the industrial policy initiatives described in the previous section still recognise this. Building domestic fabs will not be sufficient to address national security concerns when 90% of the back-end work to make semiconductors ready for installation is still done in Asia. Significant disruption in Taiwan would "leave the US with 'Made in America' chips and no devices to put them in."[55]

[55] Jan-Peter Kleinhans and Pranay Kotasthane, "Chipmaking Subsidies are Not the Answer to Supply Security Worries," Nikkei Asia, https://asia.nikkei.com/Opinion/Chipmaking-subsidies-are-not-the-answer-to-supply-security-worries; Julia Hess and Jan-Peter Kleinhans, Chip Diplomacy: Analysis of Technology Partnerships (SNV: October 2023), 7, https://www.stiftung-nv.de/sites/default/files/snv_chip_diplomacy_analysis_of_technology_partnerships.pdf.

Figure 4. Specialisation across the global supply chain today



Source: Data sourced from the Emerging Technology Observatory Supply Chain Explorer, operated by the Centre for Security and Emerging Technology.[56]

[56] "Supply Chain Explorer," Emerging Technology Observatory, Centre for Security and Emerging Technology, last modified 16 October 2022, https://chipexplorer.eto.tech/.

A hypothetical alternative with fully self-sufficient, local supply chains in each region, which met current levels of semiconductor consumption, would have required more than $1 trillion in incremental upfront investment. This would have led to a 35-65% overall increase in semiconductor prices, passed on in higher costs for consumer devices.[57]

Nonetheless, we have already seen how disruptions in the industry can feed inflation in the global economy and alter strategic trajectories: COVID-triggered chip shortages played an important role in alerting governments to the importance of strategic cooperation through technology partnerships.[58] This was one of the main messages of the UK's 2021 Integrated Review (IR), and subsequent 2023 IR Refresh. These documents advocated for an 'Indo-Pacific tilt': emphasising partnerships with countries in the Indo-Pacific where developments will have a disproportionate influence on supply chains, strategic stability, and norms of state behaviour.[59] The IR Refresh reinforces the need to turn the 'tilt' into a longer-term strategic footing. The importance of cultivating the UK-Korea relationship must be seen through this lens.

Another motivating factor is the desire to mediate the influence of China, leveraging the concerns of other countries in the Indo-Pacific, like Korea, regarding Chinese economic coercion, espionage, and interference. In the semiconductors space, this is especially important for the UK, because the UK's strategic footholds are especially vulnerable to Chinese industrial policy. China will compete directly with the UK's emerging compound semiconductor industry (on top of already controlling many of the critical minerals the UK relies on for that industry[60]), and their self-sufficiency drive will involve trying to design out foreign IP, including ARM, a part of the supply chain where the UK has a significant market share.[61] Furthermore, Chinese industrial policy is likely to lead to dumping of lagging-edge chips on the international market, inducing UK firms to buy more of them and increase reliance on an unreliable supplier.[62]

The UK-Korea relationship is the primary focus of this report, but to be comprehensive, the UK's approach to semiconductor supply chain resilience must ensure that the billions of pounds it is *not* dedicating to domestic manufacturing manifests in concrete collaborations

[57] Antonio Varas et al., Strengthening the Global Semiconductor Supply Chain in an Uncertain Era (Boston Consulting Group and Semiconductor Industry Association: 2021), 4.
[58] Julia Hess and Jan-Peter Kleinhans, Chip Diplomacy: Analysis of Technology Partnerships (SNV: October 2023), 3.
[59] HM Government, Integrated Review Refresh 2023: Responding to a More Contested and Volatile World (Cabinet Office: March 2023), 22, https://assets.publishing.service.gov.uk/media/641d72f45155a2000c6ad5d5/11857435_NS_IR_Refresh_2023_Supply_AllPages_Revision_7_WEB_PDF.pdf.
[60] Author interview with government participants, 9 January 2024.
[61] Author interview with academic participant, 10 January 2024.
[62] Author interview with academic participant, 10 January 2024.

with countries which *are*, including the US, EU, and India. Achieving buy-in from industry is a pre-requisite for this, and that in turn will be more likely if there is a clear overarching policy vision of how deeper semiconductor partnerships connect to other technology areas. AI relies on leading edge manufacturing taking place in Asia, which makes it a unique case study in the chips landscape.

2. The Technological and Strategic Relationship between AI and Semiconductors

2.1 The software-hardware relationship

The interplay between software and hardware in AI is essential to technological innovation. Without semiconductors, AI remains a theoretical concept, as it heavily relies on computational power to execute mathematical algorithms. Recent AI hype has not fundamentally altered this core principle; rather, it has spurred a shift in priorities towards specialised accelerators and chips tailored for AI applications.

Software has an integral role in maximising the performance and functionality of hardware components. The success of hardware firms often hinges on the integration of software interfaces that create a competitive advantage and differentiate their products in the market. For example, Nvidia's CUDA programming interface has become synonymous with GPU acceleration in AI applications.[63] Crucially, software stack and algorithmic developments often outpace hardware developments. This creates a cycle where hardware must interface with software once a critical stage of development is reached; for AI this is already the case.

On the other hand, the market power of cloud providers poses significant challenges to the AI hardware landscape. Cloud infrastructure providers are designing proprietary chips to meet growing demand for AI compute resources, potentially disrupting traditional chip design markets dominated by established players like Nvidia.

2.2 AI challenging the laws of compute

Historically, progress in computing has been fuelled by a symbiotic relationship between the software and hardware communities. However, AI has ushered in a new era heavily reliant on deep neural networks (DNNs) for data processing.[64] This shift has not been seamlessly

[63] Jai Vipra and Sarah Myers West, *Computational Power and AI* (AI Now: September 2023), 24, https://ainowinstitute.org/wp-content/uploads/2023/09/AI-Now_Computational-Power-an-AI.pdf.

[64] Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *Communications of the ACM* 60, no. 6 (May 2017), https://doi.org/10.1145/3065386; Yann LeCun, Yoshua Bengio, and Geoffrey Hinton, "Deep Learning," Nature 521 (2015): 436-444.

integrated into hardware architectures, save for a few exceptions.[65] This disconnect is evident in both processor speed and energy demands per operation, particularly problematic in edge computing scenarios where power- and bandwidth-intensive GPUs are unsustainable.[66]

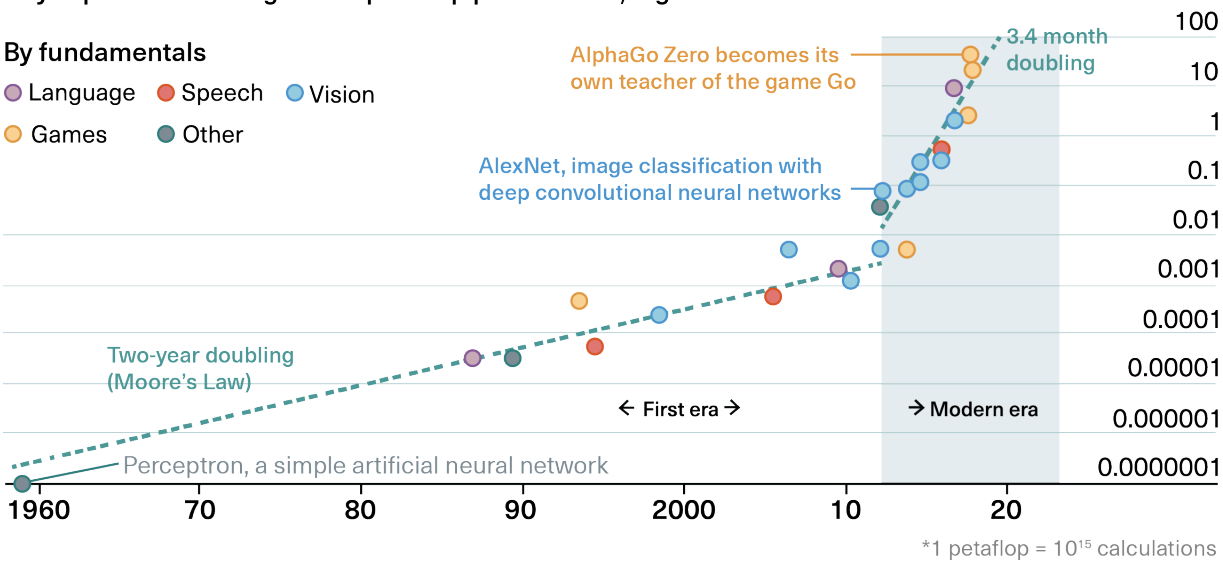
AI requires computations that are fundamentally easy for biological brains, but difficult for digital computers (i.e. image recognition tasks). All computations have thus far followed a von-Neumann architecture, which separates the actual processing from the memory. Such AI-related tasks are very memory intensive, thus creating a "von-Neumann Bottleneck" in the transfer of data between the processor and the memory.

The limitations of traditional computing architectures compared to biological computing models have long been recognised, dating back to the work of Carver Mead and others in the 1980s.[67] Now, the slowdown of Moore's Law and Koomey's Law,[68] combined with the exponential growth of AI-generated data, has created a pressing need for radical innovations in hardware design.[69]

[65] Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda, "Towards Spike-based Machine Intelligence with Neuromorphic Computing," Nature 575 (2019): 607-617.
[66] Gordon Wetzstein et al., "Inference in Artificial Intelligence with Deep Optics and Photonics," Nature 588, (2020): 39-47.
[67] Carver Mead, "Neuromorphic Electronic Systems," in Proceedings of the IEEE International Conference on Robotics and Automation 78, no.10 (Cincinnati, 1990), 1629-1636.
[68] Jonathan Koomey et al., "Implications of Historical Trends in the Electrical Efficiency of Computing," IEEE Annals of the History of Computing 33, no.3, (2010): 46-54.
[69] Bhavin J. Shastri et al., "Photonics for artificial intelligence and neuromorphic computing," Nature Photonics 15, no.2, (2021): 102-114; Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda, "Towards Spike-based Machine Intelligence with Neuromorphic Computing," Nature 575, (2019): 607-617.

These challenges are highlighted in the graph below.

Figure 5. The exponential increase in computational power requirements because of the rise of AI will require Moore's Law to scale 5x faster than originally predicted



Source: The Economist.[70]

In response to these challenges, there has been a surge of interest in neuro-inspired implementations.[71] Recent strides in electronic neuromorphic machines have showcased significant advantages in energy efficiency, latency reduction, and processing speed. These efforts aim to replicate key aspects of the brain's architecture, including its extensive interconnectivity, hierarchical organisation, and time-dependent synaptic processing. Notable projects such as IBM's TrueNorth,[72] Intel's Loihi, and the SpiNNaker[73] project (a British success story) have highlighted the potential of neuromorphic approaches.

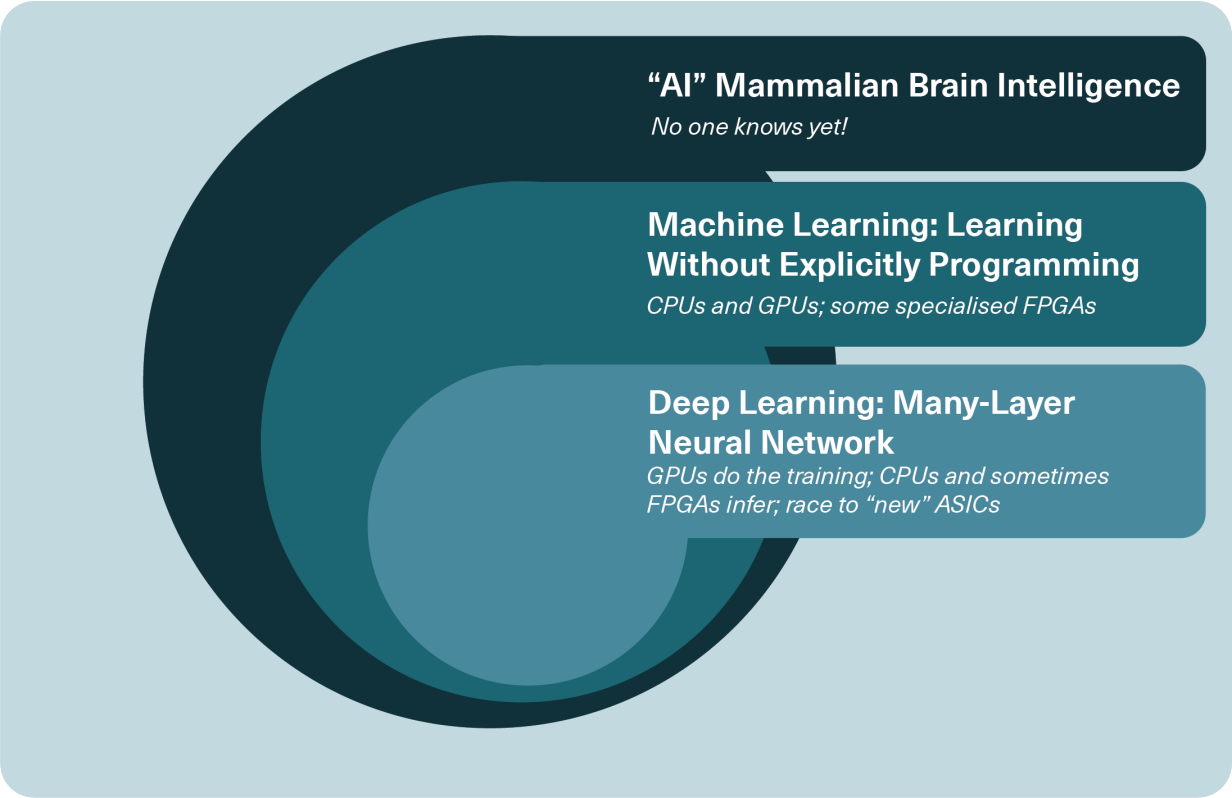
Furthermore, the emergence of photonic neuromorphic computing,[74] while still in its nascent stages, holds promise for substantial improvements in energy consumption and computational density. Operations in the optical domain offer high-speed processing, nearly lossless transmission, and massive parallelisation capabilities. Several start-ups have raised

[70] The Economist, "The Cost of Training Machines is Becoming a Problem," 11 June 2020, https://www.economist.com/technology-quarterly/2020/06/11/the-cost-of-training-machines-is-becoming-a-problem.
[71] Mike Davies et al., "Loihi: A Neuromorphic Manycore Processor with On-chip Learning," IEEE Micro 38, no.1, (2018): 82-99.
[72] Paul A. Merolla et al., "A Million Spiking-neuron Integrated Circuit with a Scalable Communication Network and Interface," Science 345, no.6197, (2014): 668-673.
[73] Steve Furber et al., "The Spinnaker Project," Proceedings of the IEEE 102, no.5, (2014): 652-665.
[74] Edwin Cartlidge, "Photonic Computing for Sale," Optica, January 2023, https://www.optica-opn.org/home/articles/volume_34/january_2023/features/photonic_computing_for_sale/.

substantial capital in this space: the UK has one photonic computing start-up using IP licensed from Salience Labs.

The rise of AI-related computation has skyrocketed demand for GPUs, which now carry out the bulk of machine learning (ML)-related tasks. GPUs conduct matrix-vector multiplication or multiply-and-accumulate (MAC) operations – this has led to its applicability in ML. To efficiently use such processors to carry out computational tasks, an application programming interface (API) is necessary and currently all GPUs use Nvidia's CUDA platform, setting a high bar for new hardware technology to make inroads. While CUDA is currently the dominant platform, new hardware research is underway, and interfaces will continue to require refinement to reflect developments in hardware.

Figure 6. AI is a catch-all term that includes machine learning and deep learning tasks; current hardware used for these applications



Source: Abu Sebastian et al., 2018.[75]

[75] Abu Sebastian et al., "Tutorial: Brain-inspired Computing Using Phase-change Memory Devices," Journal of Applied Physics 124, no.11 (2018).

One can thus summarise the rapid advance of these technologies over time as follows:

Moore's Law: See *Glossary.*

Advanced Transistor Designs: Traditional planar transistors have given way to advanced transistor designs like FinFET (Fin Field-Effect Transistor) and GAAFET (Gate-All-Around Field-Effect Transistor). These designs offer better control over the flow of electrons, leading to improved energy efficiency and performance. FinFET has become the standard for modern semiconductor manufacturing processes.

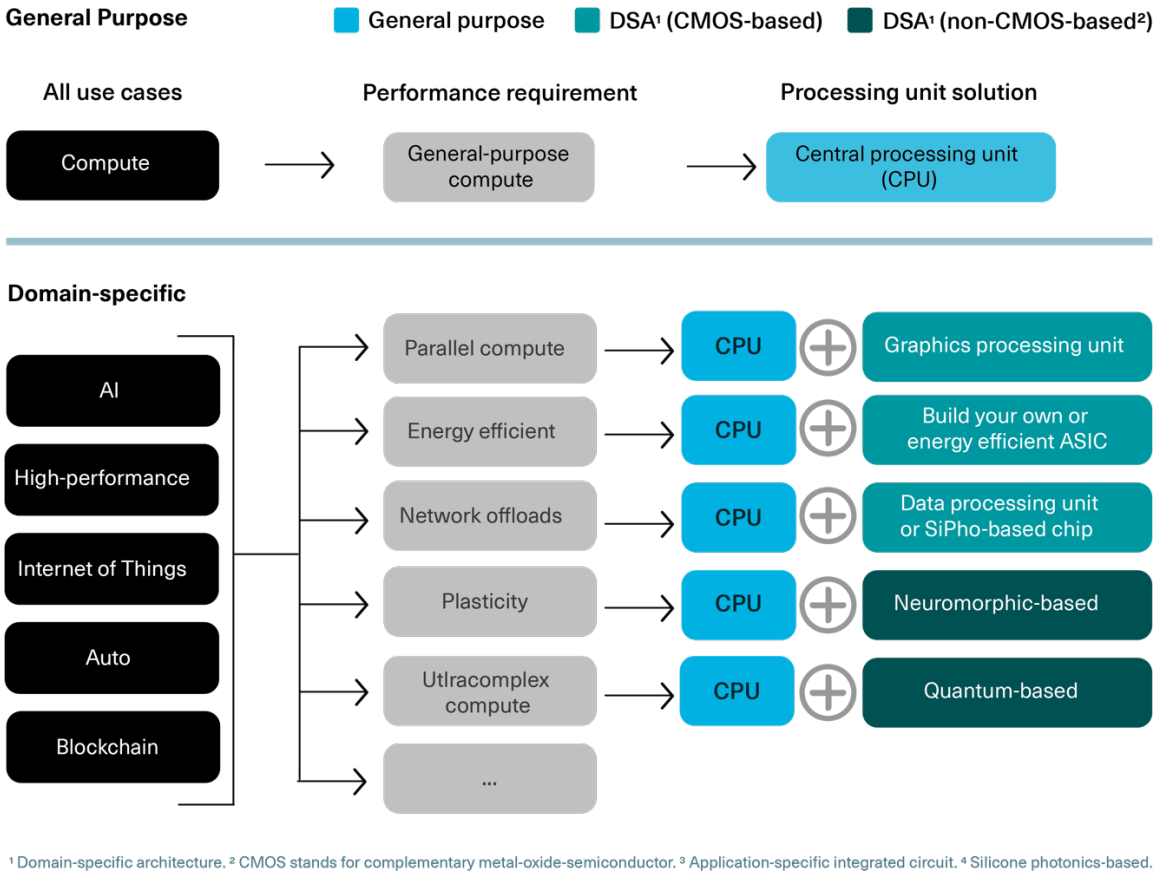
Memory Technologies: High Bandwidth Memory (HBM) and Hybrid Memory Cube (HMC) technologies offer increased memory bandwidth and density, potentially allowing AI accelerators to process large datasets.

Advanced Packaging Techniques: Traditional chip packaging methods are being replaced by advanced packaging techniques such as 3D stacking (vertically stacking multiple layers of chips) and chiplets (see *Glossary*) which are offering greater flexibility and scalability in AI hardware design.

Specialised Accelerators: AI hardware is synonymous with accelerator chips (i.e. GPUs, FPGAs) which have been optimised for tasks such as matrix multiplication, convolutional neural networks (CNNs), and recurrent neural networks (RNNs). These deliver orders of magnitude improvement in performance and energy efficiency compared to general-purpose processors.

The synergy of these semiconductor technologies has revolutionised the AI landscape. As the world moves away from a general-purpose computer architecture to more specialised compute, there will be opportunities for niche hardware companies in the AI domain. A recent report by McKinsey highlighted this transition.

Figure 7. Domain-specific architectures complement or substitute general-purpose compute by offering workload- and application-specific features



Source: McKinsey & Company, 2023.[76]

2.3 Strategic AI-semiconductor partnerships and collaborations

The convergence of AI and semiconductor technologies has spurred a wave of strategic partnerships and collaborations. The best partnerships combine expertise of AI companies in algorithm development and software optimisation with semiconductor companies' capabilities to design and manufacture at cutting-edge process nodes (Korean collaborations of this nature are discussed in Section 4).

[76] Mena Issler et al., "Domain-specific Architectures and the Future of Compute", McKinsey & Company, 18 July 2023, https://www.mckinsey.com/industries/industrials-and-electronics/our-insights/domain-specific-architectures-and-the-future-of-compute.

- **Intel and NVIDIA**: Intel is a leading semiconductor manufacturer, while Nvidia is a pioneer in GPU computing. They have indicated possible collaboration on "Confidential Computing for AI"[77] which would signal greater integration of platforms, although details remain scarce.

- **Google and TSMC**: Google, a key player in AI research and development, has a manufacturing partnership with Taiwan Semiconductor Manufacturing Company (TSMC), the world's largest dedicated semiconductor foundry, to develop custom AI chips. Google's Tensor Processing Units (TPUs), designed specifically for deep learning tasks, will be manufactured by TSMC in 2025 using advanced process nodes. These chips are used for so-called "edge" compute to be deployed in smartphones.[78]

- **Apple and TSMC**: Apple, renowned for its integration of AI technologies in products like Siri and Face ID, collaborates closely with TSMC to produce custom-designed chips, including the A-series processors used in iPhones. TSMC's advanced manufacturing processes enable Apple to achieve significant performance and energy efficiency gains. As Apple moved away from Intel-designed chips to design their own chips ("Apple silicon"), they partly drove TSMC's rise as the world's largest manufacturer.

- **AI start-ups and semiconductor foundries**: Many AI start-ups often collaborate with semiconductor foundries to design and manufacture specialised AI chips. Foundries like TSMC, GlobalFoundries, and Samsung Foundry offer access to advanced process technologies and design services, enabling start-ups to bring their AI innovations to market faster and more cost-effectively.

ARM on the global stage: partnerships with TSMC, Intel and Samsung

ARM's chips are often optimised for TSMC's advanced manufacturing processes, enabling ARM-based chips to achieve high performance and energy efficiency. This partnership plays a crucial role in driving innovation in the mobile and Internet of Things (IoT) markets. Meanwhile, Intel fabricates ARM-based chips for some customers through its foundry services division, offering a broader range of options for developing custom systems-on-a-chip (SoCs) tailored to specific use cases.

[77] InsideHPC, "Intel, Nvidia to Collaborate on Confidential Computing for AI Workloads," 7 July 2023, https://insidehpc.com/2023/07/intel-nvidia-to-collaborate-on-confidential-computing-for-ai-workloads/.
[78] Mavis Tsai and Judy Lin, "Google's Customised Tensor Chip Reportedly to be Produced by TSMC in 2025," DigiTimes Asia, 7 July 2023, https://www.digitimes.com/news/a20230707PD207/google-pixel-chips+components-tsmc.html.

The relationship between ARM and Samsung is multifaceted, encompassing collaboration, competition, and strategic partnership.[79]

- *Licensing and collaboration:* Samsung has been a significant licensee of ARM's processor IP, using ARM architectures in its Exynos line of processors for smartphones, tablets, and other devices. Samsung also works closely with ARM during the development of new ARM cores.
- *Competition in the semiconductor market:* while Samsung licenses ARM's processor designs, it also competes with ARM's licensees in certain segments of the semiconductor market.
- *Foundry services:* ARM licensees, including companies that design their own ARM-based chips, often utilise Samsung Foundry's manufacturing services to produce their silicon.
- *Strategic partnerships and alliances:* Samsung and ARM have conducted joint R&D, made technology sharing agreements and mutual investments in areas such as smartphone chips, AI, IoT, and next-generation computing.

As the AI landscape continues to evolve, we can expect to see deeper integration and collaboration between AI and semiconductor companies, leading to even more ground-breaking advancements (and concentration of market dominance among a small number of companies).

[79] Design & Reuse, "Arm and Samsung Electronics Extend Strategic Alliance with Long-term Licensing Agreement," 15 July 2002, https://www.design-reuse.com/news/3547/arm-samsung-electronics-extend-alliance-long-term-licensing-agreement.html.

2.4 Looking ahead: A 'frontier AI' dominated landscape?

The relationship between frontier AI[80] models and the semiconductor ecosystem could shape the future of both industries in the following ways:

1. **Evolution of AI accelerators:** as AI models become more sophisticated, there is a growing demand for specialised chips to handle AI workloads. Major semiconductor manufacturers are expected to develop AI accelerators, poised to compete with GPUs for market share.

2. **Critical role of semiconductors in AI advancement:** semiconductor innovation has enabled further development and proliferation of AI technologies. The demand for high-performance chips continues to rise as AI applications become more data intensive.

3. **Technological advancements:** HBM chips are crucial components in AI systems, especially for training large language models (LLMs). Companies like SK Hynix and Samsung Electronics are scaling up production of HBM chips to meet the growing demand from AI applications. An emerging trend here is in-memory compute, a technology primarily espoused by IBM[81] and emerging photonic computing platforms.[82]

4. **Compute infrastructure challenges:** the demand for high-performance chips, such as Nvidia's H100 and A100, has led to unconventional arrangements and strategies to secure compute resources. Governments and companies are investing heavily in compute infrastructure to support AI research and development.

5. **Technological diversity and innovation:** newer chip designs, such as Graphcore's Intelligence Processing Units (IPUs), offer potential alternatives to traditional GPUs for AI applications. However, challenges exist in breaking lock-in effects and promoting technological diversity in national AI compute infrastructure.

6. **Compute intensiveness and energy efficiency:** the exponential growth in AI compute demands poses significant challenges in terms of energy efficiency and

[80] Frontier AI models refer to AI models that represent the latest advancements in AI technology, typically incorporating sophisticated architectures, large-scale training data, and advanced training techniques like deep learning and reinforcement learning.

[81] Abu Sebastian et al., "Memory Devices and Applications for in-memory Computing," Nature Nanotechnology 15, no.7 (2020): 529–544.

[82] Carlos Rios et al., "In-memory Computing on a Photonic Platform," Science Advances 5, no. 2, (2019); Johannes Feldmann et al., "Parallel Convolutional Processing Using an Integrated Photonic Tensor Core," Nature 589, no.7840 (2021): 52–58.

resource consumption. Efforts are underway to address these challenges, but the industry faces substantial hurdles as it navigates the limitations of Moore's Law.

In summary, AI requires an entire system to work in concert to deliver computational advances. This entails close interplay between the software, piping and core hardware technology, so-called System Technology Co-Optimization (STCO).

As AI continues to evolve and permeate various industries, collaboration between software developers and chip designers will be essential to driving innovation and shaping future AI architectures. Strategic partnerships between leading companies in both sectors will also dictate the direction of travel: by combining expertise in AI algorithms, software development, and hardware design, these partnerships will push the boundaries of what is possible in AI-powered applications.

3. The UK Semiconductor Industry: Punching Above its Weight?

The UK semiconductor industry has seen a hive of activity building up to the National Semiconductor Strategy and following its publication. On the international front, this involved semiconductor-related agreements signed with Japan and Korea in 2023.

To build on these agreements, the UK must double down on a clear offering in the semiconductors space. Bilateral and multilateral partnerships are proliferating at speed, and the UK cannot afford to be left behind. If it is unwilling to leverage industrial policy in the same way as some other countries, it must be even more aggressive in leveraging existing semiconductor strengths, and adjacent domains of expertise in areas like AI and cybersecurity.

3.1 Recent developments

In August 2023, then Technology Minister, Paul Scully MP, said that the UK will not "recreate Taiwan in South Wales" and will not join the race across Asia, Europe and the US spending billions on advanced chip-building facilities.[83] Instead, as per the National Semiconductor Strategy[84] published in May 2023, the UK would invest up to £200m in the semiconductor sector between 2023-25, and up to £1bn in the next decade, focusing on the following three objectives:

1) **Grow the domestic sector** by establishing the Semiconductor Advisory Panel; ensuring that applied research is oriented to industry needs; improving access to infrastructure; and better attracting talent.

2) **Mitigate supply chain disruptions** by helping industry better understand risks from future shortages; improving resilience in critical sector product manufacturers; working with international partners to prioritise access to chips; and diversifying supply.

[83] Cristina Criddle, "UK Chip Sector Cannot 'Recreate Taiwan in South Wales', Says Tech Minister," Financial Times, 3 August 2023, https://www.ft.com/content/7bb96cc9-eabe-43d8-a552-2107354c3ea0.
[84] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 7, https://assets.publishing.service.gov.uk/media/646626780b72d3001334476d/national_semiconductor_strategy.pdf.

3) **Protect UK national security** by utilising investment screening and export controls; and leveraging existing hardware strengths so that cybersecurity is prioritised at the chip design stage across the world.

The total £1bn funding will support the 'UK Semiconductor Infrastructure Initiative', with some indications that this could morph into a UK National Semiconductor Institute.[85] A central coordinating function for the UK semiconductor industry was seen as sorely lacking in the current landscape by some interviewees, and it is imperative that a new National Semiconductor Institute is established this year.[86]

UK fabs do not produce the most advanced silicon semiconductors. The lack of investment into building semiconductor facilities in recent decades has led to specialised clusters forming around certain universities. The benefits of this will be highlighted later, but it also means that each fab has its own manufacturing method and end-product, with knowledge, skills, and manufacturing equipment less likely to be cross-purpose.[87]

It also means that there may be a mismatch between the output from UK fabs and the requirements of UK manufacturing or technology firms which rely on semiconductors. Further upstream, it is generally a struggle for small companies and research organisations to access the infrastructure needed to develop designs or manufacture prototypes to attract investment (although the reasons for these struggles usually vary across companies).[88]

International partnerships are increasingly being recognised as an important route to addressing domestic shortcomings and reinforcing UK strengths. The UK-Japan Digital Partnership coincided with the launch of the National Semiconductor Strategy, and there is more focus across government on identifying 'Tier 1' countries with which to prioritise bespoke agreements, also seen in the UK-Republic of Korea framework for semiconductor cooperation announced in November 2023.[89] Yet given geographical disparities, time zone differences, and more limited day-to-day cultural engagement with partners in Asia, UK stakeholders need to continually nurture these relationships in order for them to bear fruit.

Fundamentally, future UK semiconductor resilience will be dependent on the success of industrial policy in countries like the US, Korea, Japan and the EU: in the case of a China/Taiwan crisis, the UK manufacturing base will need to be able to draw on chips made

[85] Institute for Manufacturing, "UK Semiconductor Infrastructure Initiative Feasibility Study: Summary of Findings," Roadshow Presentation, 31 October 2023, 29, https://www.ifm.eng.cam.ac.uk/uploads/Roadshow_presentation_311023.pdf.
[86] Author interview with industry participant, 30 January 2024.
[87] House of Commons, The Semiconductor Industry in the UK (Business, Energy and Industrial Strategy Committee: November 2022), 12, https://committees.parliament.uk/publications/31752/documents/178214/default/.
[88] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 7.
[89] Author interview with government participants, 9 January 2024.

in allied countries, or otherwise face the prospect of seizing up entirely.[90] This is why it is important for the UK to invest in the design capability that can exploit the available friend-shored manufacturing capability, coupled with strategic agreements for reserve chip capacity in partners' fabs.

Nonetheless, there was also a view from some interviewees that the UK's relative financial prudence could, perhaps paradoxically, end up working in its favour as US-China trade disputes escalate: "this might give the UK a mild competitive edge because we are not going hot and cold on the money – simplicity could be working to our advantage."[91]

Because of semiconductors' position at the bottom of the tech stack and their facilitatory nature for other technologies, it is also informative to consider how the semiconductor industry is impacted by a wide range of 'adjacent' government strategies, reviews, and associated funding pledges, separate to the National Semiconductor Strategy. One of the main upshots of this is that there is a large amount of money in the telecoms, quantum, and AI industries that could be better aligned – making these strategies and reviews fit together as part of a coherent whole will underpin UK strategic advantages in the long run.

Table 2. Adjacent UK strategies and investments

Strategy/Review	Associated funding/pledge
Independent Review of the Future of Compute, March 2023	£900m to build a supercomputer in support of UK-based AI research and for a separate AI Research Resource (AIRR); £100m in direct GPU purchases
National Quantum Strategy, March 2023	Investing £2.5bn in quantum over the next 10 years
AI Safety Institute, November 2023	£100m annually over the next decade, subject to continued requirement
Advanced Manufacturing Plan, November 2023	'Making available' £4.5bn to support strategic manufacturing sectors over 5 years from 2025
Critical Imports and Supply Chains Strategy, January 2024	No specific funding pledge

[90] Author interview with academic participant, 10 January 2024.

[91] Author interview with government participants, 9 January 2024.

3.2 Areas of strength

The UK's National Semiconductor Strategy emphasises that 'building on our strengths will help us achieve our ambitions elsewhere: to lead the way on AI, to enable advances in quantum computing and telecommunications, to power high performance computing, and to facilitate progress towards net zero and in life sciences'.[92] It is therefore worth profiling these existing strengths and some of the barriers to further consolidation.

3.2.1 Core IP and chip design

Core IP consists of reusable, modular design blocks that chip design firms license for use in their designs. This represents the main part of the supply chain where the UK is close to market leadership. According to the Emerging Technology Observatory Supply Chain Explorer, the UK has 43% of global market share in core IP, largely due to ARM being the top core IP vendor, providing an instruction set architecture (ISA) and underpinning most of the world's smartphone processors.[93]

Section 2 touched on the centrality of ARM to UK semiconductor competitiveness, and this was also underscored by interviewees. One commented, "we must have a seat at the table. If we lose our position in the design of chips, via the likes of ARM, we become irrelevant in terms of working with the big three global companies (TSMC, Samsung, Intel)."[94] It is therefore easy to see why the eventual failure of the Nvidia-ARM merger was met with relief in UK policy and industry circles. The prospect of ARM being restricted to developing on-chip AI functions tied to Nvidia's proprietary hardware, along with the physical relocation of ARM's headquarters to the US, would have significantly undermined UK leverage in the sector.[95]

Nonetheless, ARM is not the whole story – global companies like Apple and Intel also have UK offices focused on semiconductor design and GPU development, while Graphcore are a Bristol-based company which specialise in designing hardware for AI.[96] Yet Graphcore arguably offers an example of where the government's strategic approach falls short. While

[92] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 2.
[93] "Supply Chain Explorer," Emerging Technology Observatory, Centre for Security and Emerging Technology, last modified 16 October 2022, https://chipexplorer.eto.tech/.
[94] Author interview with industry participant, 19 January 2024.
[95] United States Federal Trade Commission, "Statement Regarding Termination of Nvidia Corp.'s Attempted Acquisition of Arm Ltd.," 14 February 2022, https://www.ftc.gov/news-events/news/press-releases/2022/02/statement-regarding-termination-nvidia-corps-attempted-acquisition-arm-ltd.
[96] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 23.

Nvidia dominates the GPU market, Graphcore focuses on IPUs which are designed to be better suited to the specific requirements of AI. Because IPUs are so different to the Nvidia GPUs familiar to users, gaining a foothold in the market is a tall order. This was made even more challenging when the UK Government's Exascale funding explicitly specified GPUs, excluding IPUs from the tender.[97] The fact that the US Department of Energy's National Labs have made IPUs part of their infrastructure makes this decision even more surprising – forsaking technological diversity in the national AI compute infrastructure will further entrench applications that suit GPUs and limit techniques made possible by new AI systems.[98]

The UK Government could also be doing more to address the high costs of EDA tools which restrict production timelines for new designs and smaller companies' ability to scale up. These costs often result in companies outsourcing and missing out on commercial opportunities.[99] One promising solution that has been recommended in this regard is a government-sponsored Design Competence Centre offering design flow support, design services and EDA/IP training.[100] Working with international partners like Korea on open-source EDA tools would also be a positive way forward.

A longer-term threat to the UK's chip design strengths comes in the form of growing competitiveness of Chinese chip design companies, which have not received as much media attention as Chinese foundries (like SMIC) and memory chip manufacturers (like YMTC). In areas not targeted by US export controls, China is likely to strengthen its design capabilities, benefiting from their strength in adjacent industries like consumer electronics and EVs.[101] This symbiotic relationship between design and end-customer industries could create new dependencies on Chinese chip design, which is especially concerning given that Chinese chip design capabilities could have defence and espionage-related implications.[102] The upshot of this is that the UK must retain sovereign control as a design authority, while

[97] Peter Guest, "Graphcore Was the UK's AI Champion – Now It's Scrambling to Survive," Wired, 6 October 2023, https://www.wired.co.uk/article/graphcore-uk-ai-champion-now-its-scrambling-to-stay-afloat.
[98] Peter Guest, "Graphcore Was the UK's AI Champion – Now It's Scrambling to Survive," Wired, 6 October 2023.
[99] Institute for Manufacturing, "UK Semiconductor Infrastructure Initiative Feasibility Study: Summary of Findings," Roadshow Presentation, 31 October 2023, 25.
[100] Institute for Manufacturing, "UK Semiconductor Infrastructure Initiative Feasibility Study: Summary of Findings," Roadshow Presentation, 31 October 2023, 26.
[101] Jan-Peter Kleinhans and John Lee, "Challenges of a Rising Chinese Chip Design Ecosystem," in Europe's Strategic Technology Autonomy From China: Assessing Foundational and Emerging Technologies, ed. Tim Ruhlig (Digital Power China: A European Research Consortium, January 2023), 27-40.
[102] Jan-Peter Kleinhans and John Lee, "Challenges of a Rising Chinese Chip Design Ecosystem," in Europe's Strategic Technology Autonomy From China: Assessing Foundational and Emerging Technologies, ed. Tim Ruhlig (Digital Power China: A European Research Consortium, January 2023), 27-40.

also maintaining the capability to use the complete global design and manufacturing supply chain in service of national security.

3.2.2 Compound semiconductors

Similar concerns about China potentially undercutting UK strengths are relevant to compound semiconductors. These are semiconductor materials made from two or more elements, for example, silicon and carbon resulting in silicon carbide. South Wales is home to the CSconnected cluster, the world's only dedicated compound semiconductor cluster.[103] Although currently only 20% of chips produced are compound semiconductors (the remaining 80% are pure silicon), they do possess three properties that outperform silicon:[104]

- **Power** – i.e. electronics for EVs
- **Speed** – i.e. radio frequency for 5G and RADAR
- **Light** – i.e. photonics for optical fibre communications

The interest in compound semiconductors is partly driven by the limits of continued transistor scaling and the need for performance gains through novel technologies. There is a clear logic to the UK focusing on new architectures and materials, given that it is not a winner in the current silicon-dominated paradigm.[105] There are high value areas of the compound semiconductor industry where giants like TSMC are not active: including silicon prototyping, photonics, quantum chips and silicon carbide.[106] Indeed, one interviewee commented that "even Taiwan sit up when we talk to them about compound semiconductors, Japan too. That could be a strength for us to play internationally."[107] Samsung are showing interest in building out their compound semiconductor line, and the UK should be leveraging its strengths here to ensure that any prospective plant is built in the UK rather than other options in Europe, such as Germany.

Another driving factor are the numerous promising applications of compound semiconductors in the defence and security context. For example, indium phosphide chips are used for facial recognition and high-speed optical communications, particularly in demanding sub-sea environments, while gallium nitride and gallium arsenide play vital roles

[103] "Compound Semiconductors and Applications in South Wales," UK Department for Business and Trade, last accessed 4 April 2024, https://www.great.gov.uk/international/content/investment/opportunities/compound-semiconductors-and-applications-in-south-wales/.
[104] Louis Carney, The UK Semiconductor Industry: Current Landscape and Future Opportunities (Manufacturing Technology Centre: May 2022), 37, https://www.the-mtc.org/media/zthf3zav/semiconductors-research-paper-a4-flipbook.pdf.
[105] Author interview with academic participant, 10 January 2024.
[106] Dan Robinson, "The UK Government? On the Right Track with Its Semiconductor Strategy?" The Register, 31 October 2023, https://www.theregister.com/2023/10/31/uk_semiconductor_strategy/.
[107] Author interview with industry participant, 30 January 2024.

in devices with lasers, LIDAR, and sensors, helping connectivity and enabling use in satellite applications.[108]

Recent investments indicate a vote of confidence in UK companies' potential. The UK Infrastructure Bank co-invested £182m in Pragmatic Semiconductor[109] at the end of 2023.[110] Theoretically, the approach adopted by Pragmatic could result in fabs which are 10-100 times cheaper than a standard manufacturing facility, taking only 48 hours to go from the start of the process to a finished wafer product.[111] This encapsulates a trend of more diversification in how semiconductors are made and what they are used for.

Despite strengths in compound semiconductor R&D, there is still a recognised gap in the subsequent route to commercialisation.[112] This is why the Institute for Manufacturing – following a DSIT-commissioned study – called for an uplift in compound open-access foundry capability, proposing intervention options ranging between £100m and £250m in cost.[113]

There is currently limited overlap between the UK's research strengths in compound semiconductors and commercial ambitions in AI. Building advanced logic chips using compound semiconductor capability is still likely to be 10-15 years away, although there may still be features of an *AI system* (such as image sensors) where compound semiconductors could have a role. The UK compound semiconductor industry is showing great promise, but the reality is that it comprises a broad mixture of processes that build very different chips, so will not be a silver-bullet for any single application. More strategic alignment with other UK strengths moving forward will be key in attracting larger investment and proving economic value.

3.3 Hardware and cybersecurity

A resilient semiconductor industry is largely dependent on security throughout the supply chain. This requires designing mitigations into chip devices to protect against both supply

[108] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 18.
[109] Pragmatic Semiconductor manufactures flexible integrated circuits, using substrates made of polymers rather than silicon.
[110] Tim Bradshaw, "UK Backs Chipmaker Pragmatic to Boost Domestic Tech Manufacturing," Financial Times, 6 December 2023, https://www.ft.com/content/067ee2b6-505c-4995-8d46-cd4eb796a38b.
[111] Hannah Mullane, "Why is the world investing so much in semiconductors?" BBC News, 12 January 2024, https://www.bbc.co.uk/news/business-67576918.
[112] Author interview with industry participant, 30 January 2024; Institute for Manufacturing, "UK Semiconductor Infrastructure Initiative Feasibility Study: Summary of Findings," Roadshow Presentation, 31 October 2023, 19.
[113] Institute for Manufacturing, "UK Semiconductor Infrastructure Initiative Feasibility Study: Summary of Findings," Roadshow Presentation, 31 October 2023, 22.

chain side-channel attacks and regular operation attacks.[114] This is important because the cybersecurity risk surface is usually broadened when software interacts with hardware, rather than a vulnerability being due to software or hardware individually.[115] Vulnerabilities built into hardware may be exploited by malicious software to undermine the security of a chip and the critical applications it enables.

The previous section discussed UK chip design strengths, but from a security point of view, these strengths can be undermined once a design is sent for manufacturing, where attackers could install a backdoor or cause deliberate failure in the device.[116] New functionality added at the packaging stage can change the workings of a chip – if flaws are introduced into enough computers, then backdoors could compromise whole networks in a single attack.[117] Moreover, the threat increases as chip production and packaging gets more complex – Chinese packaging innovations in chiplets and heterogenous integration layered on top of a UK chip design could amplify security risk significantly.[118]

These risks may also be inadvertent, as with the 'Meltdown' and 'Spectre' vulnerabilities that hit microprocessors in 2018, which allowed attackers to exfiltrate data. But they reflect the need for engineering design practices to identify code or IP that has not been part of the original design specification.[119] Side-channel and fault-injection attacks also allow for chips to be targeted in the field – the type of attacks described above cannot typically be patched immediately like pure software vulnerabilities.[120]

AI also adds an expanded attack surface through interventions in the model training and deployment chain, such as trojan attack insertion into the training loop and malicious model deployment. An AI application running on an AI chip thus has an extended attack surface over and above something that does not use trained models.

---

[114] Louis Carney, The UK Semiconductor Industry: Current Landscape and Future Opportunities (Manufacturing Technology Centre: May 2022), 52.
[115] Author interview with government participants, 29 January 2024.
[116] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 52; Brady Helwig et al., National Action Plan for United States Leadership in Advanced Compute & Microelectronics (Special Competitive Studies Project: November 2023), 9.
[117] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 52.
[118] Jan-Peter Kleinhans and John Lee, "Challenges of a Rising Chinese Chip Design Ecosystem," in Europe's Strategic Technology Autonomy From China: Assessing Foundational and Emerging Technologies, ed. Tim Ruhlig (Digital Power China: A European Research Consortium, January 2023), 27-40.
[119] Louis Carney, The UK Semiconductor Industry: Current Landscape and Future Opportunities (Manufacturing Technology Centre: May 2022), 52.
[120] Brady Helwig et al., National Action Plan for United States Leadership in Advanced Compute & Microelectronics (Special Competitive Studies Project: November 2023), 10.

TrailofBits 'LeftoverLocals' disclosure

In January 2024, TrailofBits disclosed a vulnerability which highlighted the unknown security risks that persist across the ML development stack. 'LeftoverLocals' is a vulnerability that allows the recovery of data from GPU local memory created by another process on Apple, Qualcomm, AMD, and Imagination GPUs. This was of particular significance to large language and ML models run on impacted GPU platforms – TrailofBits built a proof of concept where an attacker could listen into another user's interactive LLM session.[121]

Because GPUs were initially developed for graphics computations, historically security issues have not been prioritised, meaning that GPU hardware and software stacks have iterated rapidly.[122] Several GPU frameworks do not sufficiently isolate memory in the same way expected of CPU-based frameworks, allowing an attacker with access to a shared GPU through its programmable interface to steal memory from other users and processes. This is particularly relevant to ML systems which use local memory to store model inputs, outputs, and weights.[123]

The AI chip market is booming but there is a clear risk it does so at a pace where security measures cannot keep up: GPU compute environments must be thoroughly evaluated when used for processing any sensitive data.

When interviewees were asked about plausible worst-case scenarios, many spoke of "kill switches" implanted by adversaries on circuit boards in critical infrastructure, although the likelihood of this happening at scale is very hard to determine. The limited number of processor designs and their ubiquity across most embedded devices was seen as a critical risk factor:

> 'One generic processor design could be used in the brake controller of every car in the world (...) if you can get into every car in the country, turn them all left or stop

[121] Tyler Sorensen and Heidy Khlaaf, "LeftoverLocals: Listening to LLM Responses Through Leaked GPU Local Memory," TrailofBits, 16 January 2024, https://blog.trailofbits.com/2024/01/16/leftoverlocals-listening-to-llm-responses-through-leaked-gpu-local-memory/.
[122] Tyler Sorensen and Heidy Khlaaf, "LeftoverLocals: Listening to LLM Responses Through Leaked GPU Local Memory," TrailofBits, 16 January 2024.
[123] Tyler Sorensen and Heidy Khlaaf, "LeftoverLocals: Listening to LLM Responses Through Leaked GPU Local Memory," TrailofBits, 16 January 2024.

> them all at once, that would be disruptive. The example is just illustrative but is theoretically possible because of the ubiquity of a small number of products.'[124]

This reinforces the challenges the UK faces meeting sensitive semiconductor requirements.[125] The National Semiconductor Strategy commits to 'evaluate the UK's future domestic semiconductor manufacturing needs to see where the baseline level of manufacturing could provide a low volume of chips for critical infrastructure.'[126] In the immediate term, focusing on technical solutions for hardware verification and security may be more useful due to the difficulty of avoiding reliance on overseas suppliers.

There are also numerous *market incentives* challenges which are inhibiting better security outcomes. The National Semiconductor Strategy references this, saying that chip security is 'not seen as an independent end goal' for designers and manufacturers,[127] and this was reaffirmed by interviewees:

> 'People buy capabilities because they are fast or shiny (...) often there is the expertise to help a company fix a problem, but the cost and the cascading effects of fixing it would mean they go out of business.'[128]

Building a secure chip needs to be the default option for the market, and there are lessons to be learned from recent history:

> '20 years ago, multicore processors basically had no security, and now a CPU is pretty good security wise (...) now that we have had the AI explosion, developers cannot get away with not thinking about security, we cannot wait 20 years for security to catch up.'[129]

The challenge for the UK is to leverage its status as a global authority on hardware and cybersecurity. The Digital Security by Design (DSBD) programme is a public-private partnership which aims to ensure memory safety of software without rewriting trillions of lines of code, and block cyberattacks from over 70% of today's ongoing vulnerabilities.[130]

---

[124] Author interview with government participants, 29 January 2024.
[125] Zahid Durrani and Mervyn Jones, Response to Inquiry on the Semiconductor Industry in the UK (Imperial College London: 2022), 5.
[126] House of Commons, The Semiconductor Industry in the UK: Further Government Response to the BEIS Committee's Fifth Report of Session 2022-23 (Business and Trade Committee: May 2023), 8, https://committees.parliament.uk/publications/40140/documents/195696/default/.
[127] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 53.
[128] Author interview with government participants, 29 January 2024.
[129] Author interview with government participants, 29 January 2024.
[130] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 53.

The programme has already delivered DSBD technology through ARM's Morello project, which has boosted the UK's credentials in this domain according to overseas experts:

> 'When we talk about R&D collaboration on secure computing architectures, the UK are a partner of note (...) the ARM and Cambridge-run programme is a great case study, so we have to find tangible, well-scoped ways to cooperate around particular challenges.'[131]

The essential IP to adopt the DSBD approach is freely available, allowing it to become a standard feature in chips worldwide,[132] but there is more to be done to further expand international outreach and ensure that countries like Korea are fully engaged in these endeavours as well as the US. This was also recognised by UK-based interviewees who said that the UK's DSBD and cybersecurity leverage has still not been used to full effect globally, though some also highlighted that the Morello project is only useful for some side-channel attacks, not all.[133]

Nonetheless, regarding Korea specifically, one interviewee believed this was a particularly fruitful avenue of collaboration, because of recent alarm in the country regarding cyber preparedness, referencing recent incidents such as data centre fires (heavily impacting NAVER and Kakao services) and government IT shutdowns.[134] The UK has recognised competency in this space, and there is a clear opportunity to make it a stronger bargaining chip and enable deeper collaboration with trusted allies.

3.4 Leveraging UK R&D

Underpinning the areas where the UK excels is a globally competitive R&D ecosystem, which takes a keen interest in AI research, as well as deep technical research in next-generation compute architectures.

Compute to realise AI goals

A significant driving force for the UK Government's approach to semiconductors and investments in compute is the provision of hardware needed to achieve breakthroughs in

[131] Author interview with academic participants, 11 January 2024.
[132] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 54.
[133] Author interview with government participants, 29 January 2024.
[134] Author interview with academic participant, 16 January 2024; Peter Judge, "South Korea's Government to Grill Kakao After Data Center Fire Cripples Key Services," Data Center Dynamics, 17 October 2022, https://www.datacenterdynamics.com/en/news/south-koreas-government-to-grill-kakao-after-data-center-fire-cripples-key-services/.

technologies like AI and quantum.[135] Using non-state-of-the-art AI chips increases both energy consumption and training time significantly, translating directly into cost increases – Nvidia's cutting-edge H100 GPUs have a chip-to-chip interconnect bandwidth speed of 600 GB/s compared to 300 GB/s for the H800 GPUs sold to China.[136] In recent years, OpenAI and DeepMind made decisions to be invested in or acquired by Microsoft and Google respectively primarily due to compute costs.[137]

One interviewee speculated that training runs could reach Apollo Programme levels and eat up large portions of countries' GDP,[138] while another referenced industry discussions about new power stations being needed for the data centres that drive AI compute.[139] The total energy consumption by general-purpose computing continues to grow exponentially and is doubling approximately every three years, while the world's energy production is growing only linearly, by approximately 2% a year.[140]

The UK's Advanced Research and Innovation Agency (ARIA) has launched a £42m moonshot programme seeking to meet this challenge, focusing on principles found ubiquitously in nature to create more efficient options for AI infrastructure at 1/1000th the cost.[141] This is an example of ambitious R&D on next-generation compute architectures which the UK must leverage globally.

It is not just training AI models but also evaluating them which depends on reliable access to compute. This is central to the UK's ambitions to be a global leader in AI safety research and evaluation, and the recently established AI Safety Institute will receive priority access to the £300m AI research resource (AIRR).[142] These developments are especially important considering the recent UK-US partnership on science of AI safety.[143]

---

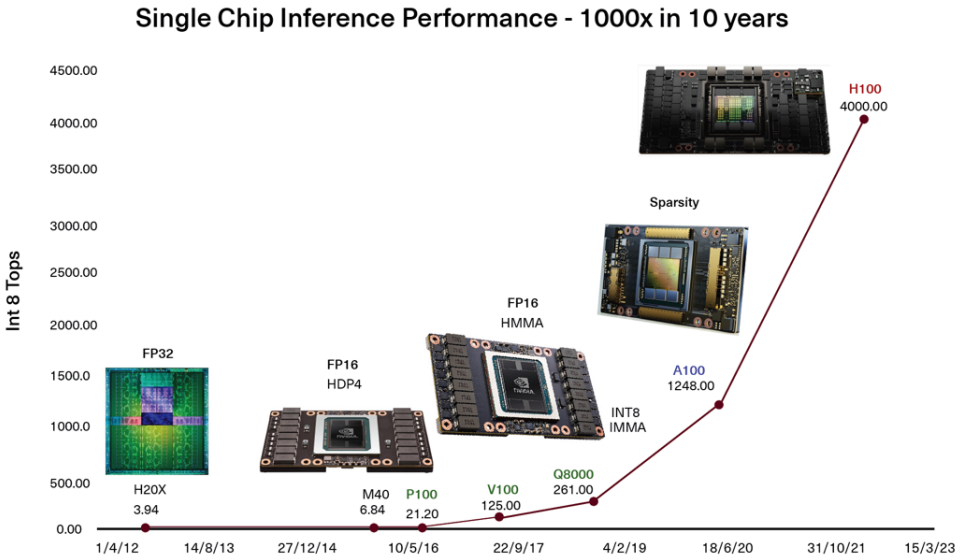
[135] HM Government, National Semiconductor Strategy (Department for Science, Innovation and Technology: May 2023), 2.
[136] Jai Vipra and Sarah Myers West, Computational Power and AI (AI Now: September 2023), 15; Stephen Nellis and Jane Lee, "Nvidia Tweaks Flagship H100 Chip For Export to China as H800," Reuters, 22 March 2023, https://www.reuters.com/technology/nvidia-tweaks-flagship-h100-chip-export-china-h800-2023-03-21/.
[137] Jai Vipra and Sarah Myers West, Computational Power and AI (AI Now: September 2023), 31.
[138] Author interview with academic participants, 11 January 2024.
[139] Author interview with industry participant, 19 January 2024.
[140] Semiconductor Research Corporation, The Decadal Plan for Semiconductors (January 2021), 123.
[141] Suraj Bramhavar, "Unlocking AI Compute Hardware at 1/1000th the Cost," *Advanced Research and Innovation Agency*, November 2023, https://www.aria.org.uk/wp-content/uploads/2023/11/ARIA-Unlocking-AI-compute-hardware-v1.0-1.pdf.
[142] "£300 Million to Launch First Phase of New AI Research Resource," UK Research and Innovation, 1 November 2023, https://www.ukri.org/news/300-million-to-launch-first-phase-of-new-ai-research-resource/.
[143] "UK & United States Announce Partnership on Science of AI Safety," HM Government, Press Release, 2 April 2024, https://www.gov.uk/government/news/uk-united-states-announce-partnership-on-science-of-ai-safety.

AI to realise chip goals

At the same time, the potential of AI to produce better chips is becoming clearer. More companies are rolling out internal AI tools to improve chip design (Nvidia has been doing so for several years), while AI is also being used to determine equipment and process failure in fabs, which can often occur unexpectedly. This is noteworthy in the context of the government's new Advanced Manufacturing Plan.[144] Additionally, generative AI may help to reduce chip design times and enable simpler natural language inputs instead of needing specialised programming languages.[145]

Algorithmic progress is a significant component of performance improvement, and has played a more important role in Nvidia achieving 1000x gains in GPU performance over the last 10 years than pure hardware efficiency (see Figure 8 below).[146] Given both the fact that UK AI strengths are more pronounced at the software/algorithm level rather than the hardware level, and the importance of the chip design sector to the UK's semiconductor positioning, there is a clear logic to the UK leaning into this dynamic more aggressively.[147]

Figure 8. Nvidia GPU improvement over 10 years



Source: Nvidia, 2023.[148]

[144] Author interview with academic participant, 10 January 2024; Author interview with academic participants, 11 January 2024.
[145] Brady Helwig et al., *National Action Plan for United States Leadership in Advanced Compute & Microelectronics* (Special Competitive Studies Project: November 2023), 27.
[146] Author interview with academic participants, 11 January 2024.
[147] Author interview with industry participant, 30 January 2024.
[148] Rick Merrit, "Heeding Huang's Law: Video Shows How Engineers Keep the Speedups Coming," Nvidia, 29 September 2023, https://blogs.nvidia.com/blog/huangs-law-dally-hot-chips/.

In 2019, South Korea's then president, Moon Jae-in, said that if the country could "link AI primarily with the sectors in which we've accumulated extensive experience and competitiveness, such as manufacturing and semiconductors, we will be able to give birth to the smartest yet most humanlike artificial intelligence."[149] For the UK, the reverse may be more instructive: utilising its world-leading research base in AI and novel forms of computing may be vital in strengthening the UK's position across the semiconductor supply chain.

[149] Government of the Republic of Korea, *National Strategy for Artificial Intelligence* (Ministry of Science and ICT: 17 December 2019), https://www.msit.go.kr/bbs/view.do?sCode=eng&nttSeqNo=9&bbsSeqNo=46&mId=10&mPid=9.

4. The South Korea Semiconductor Industry: A National Treasure

In an age of hyper-uncertainty, Korea and the UK can work closer together to secure competitive advantage in critical technologies. The semiconductor-AI lens presents an excellent opportunity to leverage the complementarities between both countries. This section focuses on the strengths that Korea brings to this equation.

4.1 Recent developments

4.1.1 Semiconductors

Korea has maintained the second largest market share in the global semiconductor industry and has been the market leader in memory semiconductors since 2000. Korea occupies 18.4% of the global semiconductor industry (56.9% of memory (DRAM/NAND), 3% of non-memory (DAO)).[150]

Yet despite considerable strengths, Korea has experienced several crises due to its high dependency on materials, parts, and equipment (MPE) and vulnerability to geopolitical risks. Three recent cases are informative here:[151]

- Japanese export controls on three key chemicals (hydrogen fluoride, photoresist, and fluorinated polyimide).
- US export controls on equipment (such as Extreme Ultraviolet Lithography (EUV)).
- Chinese export controls on gallium and germanium.

To overcome structural vulnerabilities, Korea has developed several national strategies to support semiconductor competitiveness, the most significant of which are the 'National

[150] Ji-Hoon Kim, Sungyeob Yoo, and Joo-Young Kim, "South Korea's Nationwide Effort for AI Semiconductor Industry," *Communications of the ACM 66*, no.7 (July 2023): 46.

[151] Seungjoo Lee, "High-Tech Competition in the Age of Hyper-Uncertainty: The Evolution of South Korea's Economic Security Strategy," *CETaS Expert Analysis* (May 2023); Baek Seoin et al., "US, China, and EU's National, Economic, Technological Security Strategies and Implication for Korea," Science and Technology Policy Institute, 18 August 2022; Andrew Grotto and Chaeri Park, *The Silicon Allies: Achieving Allied Resiliency Against Threats to the Semiconductor Supply Chain* (Stanford University Cyber Policy Center: May 2022), https://fsi9-prod.s3.us-west-1.amazonaws.com/s3fs-public/silicon_alliesv3_1.pdf.

Advanced Strategic Industry Act (2022)'[152] and the 'National Strategic Technology Act (2023)'.[153]

The National Advanced Strategic Industry Act features extensive financial and regulatory incentives promoting strategic industries across 12 key technology areas, as well as designating strategic industry clusters. In January 2024, Korea's Ministry of Trade, Industry, and Energy (MOTIE) and Ministry of Science and ICT (MSIT) announced a plan to build the world's largest semiconductor mega-cluster, which will have the capacity to produce 7.7 million wafers per month by 2030.[154] The Act also designates specialised research institutes, establishes regional technology innovation hubs and strengthens mission-oriented R&D through a National Critical Technology (NCT) laboratory system.

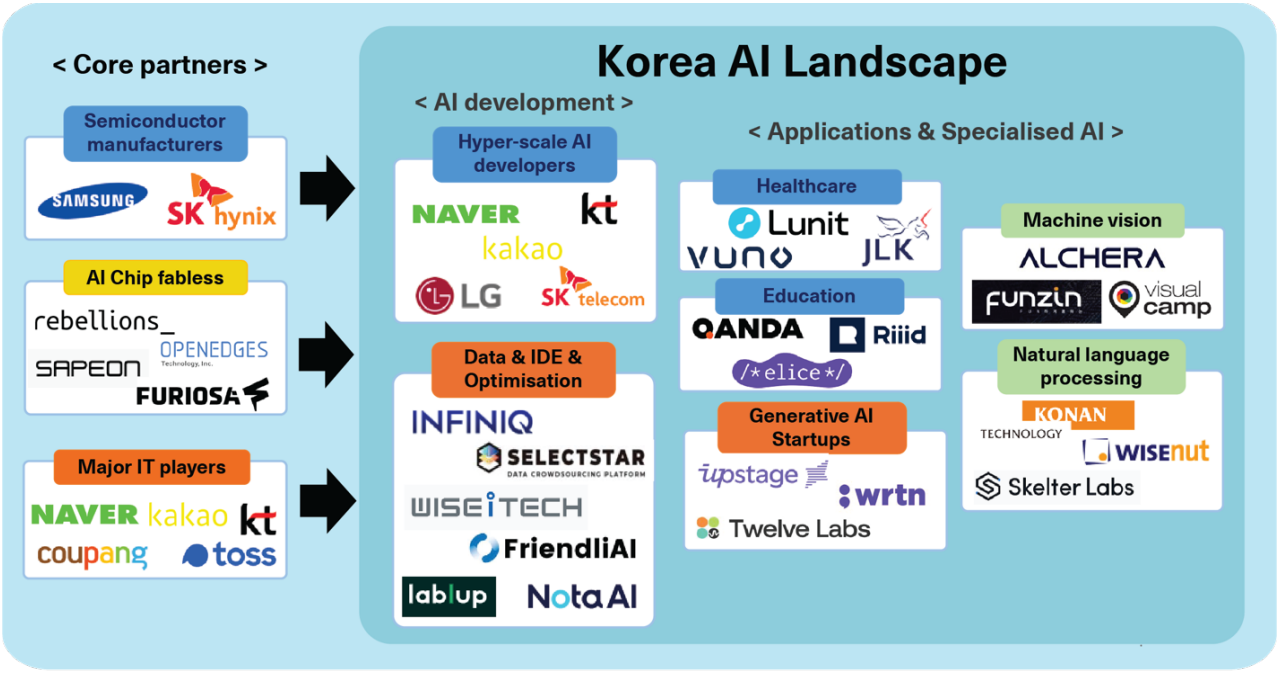
At the same time, Korea's private sector is committing large-scale investments. For example, Samsung Electronics announced its 'Semiconductor Vision 2030' which included $55 billion investment in R&D and $45 billion in infrastructure in 2019. Additionally, Samsung Electronics has initiated world-leading production based on the 3 nm foundry process incorporating Gate-All-Around (GAA) technology.[155]

4.1.2 Artificial Intelligence

The Korean AI landscape is characterised by the growing number of companies across the entire AI value chain. Korea benefits from a sovereign data ecosystem: NAVER, not Google, leads the search engine market, and domestic social media platforms also dominate.[156] The Korean Government is also supporting these data ecosystems by constructing a national integrated database called the 'Public Data Portal'.[157]

[152] "Act On Special Measures for Strengthening the Competitiveness of, and Protecting National High-tech Strategic Industries 2022" (Republic of Korea), Act No. 19205.
[153] "Special Act on the Fostering of National Strategic Technology 2023" (Republic of Korea), Act No. 19236.
[154] Government of the Republic of Korea, "Government to Establish the World's Largest and Most Advanced Mega Semiconductor Cluster," Ministry of Science and ICT, 15 January 2024, https://www.msit.go.kr/eng/bbs/view.do?sCode=eng&mId=4&bbsSeqNo=42&nttSeqNo=952.
[155] Samsung, "Samsung Begins Chip Production Using 3nm Process Technology with GAA Architecture," Samsung News, 30 June 2022, https://semiconductor.samsung.com/news-events/news/samsung-begins-chip-production-using-3nm-process-technology-with-gaa-architecture/.
[156] Colleen Chapco-Wade-Safina, "In Korea, Culture Matters in the Social Media Landscape," Medium, 27 December 2018, https://colleenchapcowadesafina.medium.com/in-korea-culture-matters-in-the-social-media-landscape-e2fad0cd8043.
[157] "Public Data Portal," Republic of Korea Ministry of the Interior and Safety, last accessed 4 April 2024, https://www.data.go.kr/en/index.do.

Figure 9. Korea AI Landscape



(c) 2024 by Hyunjin Lee, Overseas Economic Research Institute, Korea Eximbank

Source: Author, Hyunjin Lee, 2024.

Korea's AI capabilities are dispersed across several established companies, such as NAVER, LG, and KT, and promising start-ups across diverse fields like healthcare (Lunit), LLMs (Upstage), and education (Quanda). Since 2019, the Korean government has implemented a comprehensive strategy to build on this foundation, focusing on boosting competitiveness, expanding AI adoption across all sectors, and ensuring ethical implementation.[158]

[158] Government of the Republic of Korea, National Strategy for Artificial Intelligence (Ministry of Science and ICT: 17 December 2019).

Table 3. Korea AI strategies

Year	Act & Strategy	Major direction
2019	**National Strategy for Artificial Intelligence**	Promote infrastructure expansion, strategic technology development, regulatory innovation, and start-up ecosystem to boost AI competitiveness.
2020	**Artificial Intelligence Ethical Standards**	Establish ethical AI standards in line with global institutions such as the OECD, and announce roadmap for improvement of AI laws, systems, and regulations.
2021	**Plan to spread the use of AI across all regions and industries**	Empower local authorities, fostering regional collaboration, and integrate AI with local data to drive digital transformation nationally.
2022	**Korea Digital Strategy**	Implement the 'New York Initiative' announced on 21 September 2022 by President Yoon; set a goal for all citizens to be actively using AI by 2025.
2023	**Plan to strengthen the competitiveness of large-scale AI**	Boost AI application in key areas through public-private partnerships, foster 10,000 software as a service (SaaS) companies by 2026, build a massive dataset for AI development, and train 65,000 AI professionals by 2027.

Source: Summary of MSIT documents and press releases.

These efforts have already yielded tangible results, including major advancements in flagship AI projects, AI-powered healthcare systems, and nationwide AI literacy programs. Yet challenges remain in balancing innovation with ethical considerations: the National

Human Rights Commission recently argued for the elimination of the 'principle of first accept and later regulate'.

The Korean Government is proactively leveraging strengths in cutting-edge fields like semiconductors and AI in its international partnerships.[159] Collaborations are already underway with the UK, Singapore, and the US. Additionally, large-scale projects in Saudi Arabia offer promising opportunities for Korean companies, backed by government support.

Table 4. Recent international cooperation initiatives

Partner	Cooperation
 **United Kingdom**	• The Downing Street Accord announced a series of new partnership initiatives, including in technology fields like AI and semiconductors (November 2023). • Korea Institute for Advancement of Technology and Innovate UK signed an MoU for joint R&D; UK government granted £4.5 million to fund research partnerships (March 2023).
 **Singapore**	• Korea-Singapore Digital Partnership Agreement (November 2022). • Signed an AI MoU and launched a joint project with MSIT to promote policy exchanges, including joint AI research, technology exchange and licensing cooperation in the digital healthcare field, and mutual verification of AI stability and reliability.
 **United States**	• MSIT visited the US National Science Foundation (NSF), New York University, and the IBM Watson institute to discuss international research and fostering global talent on strategic technologies, including AI and semiconductors (September 2023). • The Korea-US-Japan trilateral summit was held at Camp David and agreement reached on cooperation on international AI and digital norms (August 2023).
 **Saudi Arabia**	• Korea and Saudi Arabia are pursuing large cooperation projects worth a total of $30 billion and have signed a total of 26 contracts and MoUs across various industries, including AI (November 2022). • NAVER won a $100 million digital twin platform construction project from the Saudi Arabian Ministry of Autonomy, Administration and Housing (October 2023).

[159] Government of the Republic of Korea, *The Yoon Suk Yeol Administration's National Security Strategy: Global Pivotal State for Freedom, Peace, and Prosperity* (Office of National Security: June 2023), https://overseas.mofa.go.kr/eng/brd/m_25772/view.do?seq=16&page=1.

4.2 Areas of strength

On the semiconductors front, Korea's strength is based on two of the world's top-tier IDMs – Samsung Electronics and SK Hynix. By maintaining a dominant position for over 20 years, Korea has succeeded in possessing the most advanced technology and production capacity, along with the most sophisticated ecosystem.

Furthermore, Korea's competitiveness in relatively weaker areas such as foundries is improving. In the foundry sector, Samsung Electronics is investing aggressively in its capabilities. In the fabless sector, several innovative start-ups such as Rebellions and Furiosa are growing rapidly. The active participation of a domestic digital platform/cloud provider (NAVER) possessing advanced LLM capacity also enhances Korea's competitiveness in chip design.

Meanwhile, Korea also holds impressive strengths in AI: the Global AI Index rankings place Korea 3rd in development, 6th in government strategy, and 7th in infrastructure,[160] while the Government AI Readiness Index places Korea 6th.[161] Global AI patent applications are also high (4th) although only 7% of patents fall within the top 10% of the patent citation index, falling short compared to the 14% average of the leading 10 countries.[162]

Several major Korean companies have established dedicated research organisations, including Samsung AI Center, NAVER AI Lab, Kakao Brain, and LG AI Research Institute. These companies are increasingly focused on building and commercialising their internal LLM capacity.

4.3 SK Hynix/Samsung strategic pivots

4.3.1 SK Hynix: Mission-Oriented Semiconductor-AI innovation

To achieve twin transformation, the SK Group is pursuing three strategies at the intersection of semiconductors and AI:

- Developing next-generation semiconductors (e.g. HBM) for generative AI.

[160] The Global AI Index," Tortoise Media, last modified June 2023, https://www.tortoisemedia.com/intelligence/global-ai/.
[161] "Government AI Readiness Index 2023," Oxford Insights, last modified 20 December 2023, https://oxfordinsights.com/wp-content/uploads/2023/12/2023-Government-AI-Readiness-Index-2.pdf.
[162] Clarivate and Korea Advanced Institute of Science and Technology, The Innovation Landscape of Artificial Intelligence: The Global Competition to Build a Successful AI Ecosystem (2021), https://discover.clarivate.com/AI_Innovation_Landscape_Report_download.

- Introducing AI into its core industries to enhance manufacturing and IT service competitiveness.
- Refreshing its international strategy considering heightened geopolitical risk.

Enhancing Future Technology Competitiveness with Open Innovation

SK Hynix's Revolutionary Technology Center (RTC) conducts next-generation semiconductor research, focusing on key areas such as future DRAM, NAND, new memory, quantum computing and neuromorphic computing. SK Hynix has successfully developed the 5th generation of HBM and initiated sample supply for customers. HBM vertically connects multiple DRAMs to enhance data processing speed and is recognised as a high-value, high-performance product capable of meeting the data processing needs of generative AI.[163]

SK Hynix's future technology strategy also prioritises international collaborations with prestigious institutions like the Interuniversity Microelectronics Center (IMEC), the National Institute of Standards and Technology (NIST) and the Semiconductor Research Corporation (SRC). The open, international positioning of the RTC has enabled the exploration of cutting-edge technologies and the pursuit of new approaches to overcome the limitations of existing memory semiconductors.

Two-track domain driven AI-semiconductor strategy by Gauss Labs and Sapeon

The SK Group has established two independent AI companies, Gauss Labs and Sapeon, to enhance its AI-semiconductor capacity. The former is pursuing 'AI for manufacturing innovation,' while the latter is concentrating on 'AI for IT service innovation.'

Gauss Labs uses data generated from SK Hynix plants to optimise manufacturing at several stages, including process management, yield prediction, equipment maintenance, material measurement, and defect inspection and prevention. Meanwhile, Sapeon aims to develop its own neural processing unit (NPU) and process in memory (PIM) for advanced IT services – Sapeon successfully developed and commercialised Korea's first AI chip for data centres (X220). The ongoing development of the Sapeon X300 series supports AI model training and is being developed for various fields such as automotive, security, and media.

[163] "Pioneering Excellence in 2023," SK Hynix, last modified 29 December 2023, https://news.skhynix.com/pioneering-excellence-in-2023/.

Refreshing international strategy: the geopolitical risk approach

The SK group has historically maintained strong business ties with China. However, recent geopolitical tensions have motivated the SK Group to adopt a new international strategy with a more cautious approach to operations in China. The SK Group is establishing offices in the US and investing heavily in manufacturing plants there, while also seeking closer alignment with the US government on economic security matters. This closer proximity is helping to deal with requirements for accessing CHIPS Act subsidies and to mitigate the impact of scandals like SK Hynix products being installed in Huawei's new smartphones.

4.3.2 Samsung Electronics-NAVER: AI-semiconductor collaboration for mutual competitive advantage

Global technological competition is intensifying, meaning that companies lacking a unique offering are at risk. To overcome this, NAVER and Samsung Electronics (hereafter Samsung), Korea's leaders in AI and semiconductors respectively, launched a taskforce in December 2022 to develop semiconductor solutions optimised for AI systems. The main objective was to solve AI data bottlenecks and maximise power efficiencies.

Mutual complementarity for sustainable growth

Samsung faces limitations in the fabless and foundry fields despite its dominant position in the memory field. Commercialising AI semiconductors presents a strategic move to not only strengthen its weaker presence in the non-memory market but also to secure a leadership position in an AI-driven future. Samsung wants to both reinforce existing technologies like HBM, compute express link (CXL), and PIM, and explore novel hardware structures specifically designed to accelerate AI model compression techniques.[164] These tailored structures would bring significant efficiency boosts and reduce the amount of data transferred between processing units and memory.

Developing such chips requires expertise in both semiconductor design and manufacturing and AI algorithm design and verification, with a specific focus on optimising hardware structure for specific AI architectures and model compression techniques. Therefore, collaborating with AI specialists like NAVER is crucial for Samsung's success in this new market.

[164] Samsung, "Samsung Electronics and NAVER Team Up To Develop Semiconductor Solutions Optimized for Hyperscale AI," Samsung News, 6 December 2022, https://news.samsung.com/global/samsung-electronics-and-naver-team-up-to-develop-semiconductor-solutions-optimized-for-hyperscale-ai.

On the other hand, NAVER possesses a near-monopoly on the Korean IT industry. Based on strong network effects, NAVER has accumulated large amounts of high-quality data, which has fed into their superior ML capabilities. Server costs and availability are a huge challenge to providing hyper-scale AI services,[165] which has meant several major AI companies have started developing their own AI chips. Indeed, in January 2024, OpenAI CEO Sam Altman visited Samsung and SK Hynix to discuss AI-semiconductor collaboration – this link was reaffirmed when Samsung and SK Hynix were suggested as potential partners in reports of a multi-trillion-dollar fundraising vision.[166]

Samsung's capabilities across the supply chain make them a compelling partner for many global companies seeking to develop specialised AI semiconductors. However, it is closer to home through NAVER where the strongest partnership has emerged so far.

Progress and further cooperation

The partnership works along the following lines. NAVER identifies key technical challenges for their AI services, while Samsung leverages its hardware expertise to develop solutions. NAVER then validates these solutions with their software know-how. Both companies prioritise enhancing computational and inferential capabilities while maximising power efficiency.

A year into the partnership, they unveiled their first AI-semiconductor prototype, demonstrating an 8x or more improvement in power efficiency compared to current commercial semiconductors.[167] Design is now under construction aiming for mass production by the end of 2024.[168]

Samsung has set 'hardware solutions for on-device AI' as an important goal.[169] Aligned with this vision, the Samsung-NAVER partnership is known to have adopted low power DDR DRAM instead of HBM. The chips will be manufactured at Samsung's foundry and deployed

[165] This term is used interchangeably with 'frontier AI'.
[166] Keach Hagey and Asa Fitch, "Sam Altman Seeks Trillions of Dollars to Reshape Business of Chips and AI," *Wall Street Journal*, 8 February 2024, https://www.wsj.com/tech/ai/sam-altman-seeks-trillions-of-dollars-to-reshape-business-of-chips-and-ai-89ab3db0.
[167] Jasmine Choi, "Samsung, NAVER Reveal AI Semiconductor 8x More Power Efficient than Nvidia Chips," *BusinessKorea*, 20 December 2023, https://www.businesskorea.co.kr/news/articleView.html?idxno=208150; Government of the Republic of Korea, "MSIT hosts the 4th Top-Level Strategic Dialogue on AI Semiconductors," Ministry of Science and ICT, 20 December 2023, https://www.msit.go.kr/eng/bbs/view.do;jsessionid=pJsmQ-Zxr72rRsCgyqvwPmVCaAmf3EIfa0FZ60tY.AP_msit_1?sCode=eng&mPid=2&mId=4&bbsSeqNo=42&nttSeqNo=940.
[168] ASIC is a semiconductor for specific applications, its representative products include neural network processing units (NPUs) and tensor processing units (TPUs).
[169] Samsung, "Harnessing the AI Era with Breakthrough Memory Solutions," 9 January 2024, https://semiconductor.samsung.com/news-events/tech-blog/harnessing-the-ai-era-with-breakthrough-memory-solutions/.

in NAVER's data centres. If successful, Samsung and NAVER plan to offer the chip to other global tech giants.

4.4 Leveraging semiconductor strengths for a globally competitive AI industry

The simultaneous pursuit of a domestic AI-semiconductor ecosystem and proactive international cooperation is crucial to Korea's overall approach. Korea must strengthen areas of existing competitive advantage and build buffers where it is weaker.

The Korean semiconductor industry benefits from the rise of AI chips, as evidenced by the increase in demand for SK Hynix's HBM. But mechanisms are still needed to ensure market stability and coordination. To this end, MSIT has launched the 'AI Semiconductor Scaling Network', which aims to support links between AI chip buyers, suppliers, and researchers to create a highly networked domestic ecosystem.[170] This is something that DSIT should take a close interest in and consider mirroring in the UK.

The Korean Government is also developing a more joined up approach with key semiconductor allies. One example of this was an agreement during the August 2023 trilateral US-Korea-Japan summit to launch a pilot project focusing on supply chain early warning systems.[171] The warming of relations with Japan has seen Samsung invest 40 billion yen (£210m) in an advanced semiconductor R&D centre in Yokohama, Japan, which will facilitate joint research between Korea and Japan on semiconductor packaging technology. At the same time, SK Hynix is working more closely with Taiwan, outsourcing some processes for HBM4 to TSMC.

The domestic knowhow that Korea is developing through the Samsung-NAVER partnership is positioning it as a key country in the global AI-semiconductor nexus. This is also reflected in government policy: in January 2023, the Korean Government decided to invest KRW 260 billion (£153m) in the development of next-generation AI, and in June 2023, MSIT announced an investment of KRW 1.02 trillion (£600m) to create *an industry-academia-*

[170] Government of the Republic of Korea, "AI Semiconductor Scale-up Network Launched," Ministry of Science and ICT, 30 September 2022, https://www.MSIT.go.kr/bbs/view.do?sCode=user&bbsSeqNo=94&nttSeqNo=3182205, (in Korean).
[171] The White House, "The Spirit of Camp David: Joint Statement of Japan, the Republic of Korea, and the United States," White House Statement, 18 August 2023, https://www.whitehouse.gov/briefing-room/statements-releases/2023/08/18/the-spirit-of-camp-david-joint-statement-of-japan-the-republic-of-korea-and-the-united-states/.

research institute collaboration ecosystem.[172] This brings together (1) multinational companies like Samsung and SK Hynix, which are at the core of the semiconductor-AI ecosystem, (2) start-ups such as Furiosa AI, Rebellions, Sapeon, Telechips, and DEEPX, and (3) research institutions like The Korea Advanced Institute for Science and Technology, The Electronics and Telecommunications Research Institute, and Seoul National University. The Korean government will allocate KRW 2 trillion (£1.2bn) over the next five years to support the development of AI semiconductor libraries and compilers. Moreover, data centres, which are one of the largest sources of demand for AI semiconductors, will be built to generate initial demand.

These measures could ensure Korea is better positioned to capitalise on increased global demand for AI systems outside of the US or China, in the event of an escalating trade war. The UK could occupy a similar position in this regard, potentially working closely with Korea on joint ventures in third countries.

[172] Government of the Republic of Korea, "MSIT to Announce the "Support Plan for AI Semiconductor Industry Promotion"," Ministry of Science and ICT, 27 June 2022, https://www.msit.go.kr/eng/bbs/view.do?sCode=eng&mId=4&mPid=2&pageIndex=&bbsSeqNo=42&nttSeqNo=702&searchOpt=ALL&searchTxt=.

5. Semiconductors and Geostrategic Alliances: Opportunities for the UK and Korea

Geopolitical risks are creating a dichotomy between technological competition and cooperation. On one hand, individual countries are driven to prioritise their own national interests, leading to fragmentation rather than integration of technologies and industries. On the other hand, the need for cooperation is increasing because of the importance of an open innovation ecosystem to securing competitive advantages in high-tech industries.

This section seeks to bring together the individual analyses of the semiconductor-AI landscapes in the UK and Korea, to demonstrate why the two countries are an ideal case study in bilateral technology and security partnership.

5.1 A high-point in UK-Korea relations

The November 2023 Korean State Visit to the UK coincided with the celebration of 140 years of diplomatic relations between the two countries. Last century, the UK contributed around 90,000 soldiers to fight in the Korean War; the focus today is more on finding ways to bolster mutual technological and economic security.

The State Visit, which resulted in the signing of the Downing Street Accord, was a landmark moment in formalising both countries' shared intention to strengthen and deepen bilateral collaboration across several critical sectors.[173] In a speech to the UK Parliament, President Yoon spoke of their bilateral relations being "reborn as true 'Global Strategic Partners'" and the need for them to "stand in solidarity and respond to many of the world's challenges."[174] Prime Minister Rishi Sunak said, "as two nations focused on innovation, harnessing new technologies and defending the international rules-based order, the UK and ROK are natural partners", recognising the value of the nexus between technology and global security.[175] This is also timely given that Korea is a member of the UN Security Council from 2024-25.

[173] Ardi Janjeva, "A High Point in UK-South Korea Relations? Reflecting on President Yoon's State Visit and the Path Forward for Technology and Security Cooperation," *CETaS Expert Analysis* (December 2023).
[174] UK Parliament, "President of the Republic of Korea Addresses Parliament," 21 November 2023, https://www.parliament.uk/business/news/2023/november-2023/president-of-the-republic-of-korea-addresses-parliament/.
[175] HM Government, "UK and South Korea to Agree New Partnership to Redefine and Strengthen Ties for Next Generation," Prime Minister's Office, 10 Downing Street, 20 November 2023, https://www.gov.uk/government/news/uk-and-south-korea-to-agree-new-partnership-to-redefine-and-strengthen-ties-for-next-generation.

In a recent report, Imagination Technologies laid out general criteria for an ideal UK strategic interdependence partner. These are informative for thinking about UK-Korea complementarities vis-à-vis security and technology.[176]

Table 5. Criteria for a strategic interdependence partner: Korea

Criteria	Korea suitability
Politically stable and subject to minimal external/internal threats to stability of supply to the UK.	Despite the perpetual threat from the DPRK, South Korea maintains high levels of internal political stability.
Signatory to World Trade Organisation Information Technology Agreement (ITA) and ITA expansion, ensuring a 0 tariff for semiconductors.	☑
Have an existing Free Trade Agreement with the UK.	☑
High levels of FDI in high technology manufacturing in the UK.	According to the latest data (2021), total UK inward FDI stocks stand at £1.9bn for Korea.[177] This is relatively low in comparison to Singapore (£12bn) and Japan (£92bn) but shows there is ample room for growth.
Mutual participation in collective defence agreements or strategic cooperation agreements.	Following the Korean President's State Visit to the UK in November 2023, the Downing Street Accord included a new UK-ROK Strategic Cyber Partnership; a UK-ROK Framework for Semiconductor Cooperation; a new UK-ROK Supply Chains

[176] Imagination Technologies and Global Counsel, The Future of the UK's Semiconductor Strategy (2022), 12.
[177] Office for National Statistics, Foreign Direct Investment Involving UK Companies: 2021 (23 January 2023), https://www.ons.gov.uk/economy/nationalaccounts/balanceofpayments/bulletins/foreigndirectinvestmentinvolvingukcompanies/2021.

	Dialogue; and launching the ROK-UK Digital Partnership.[178]
Track record of restraint in export restrictions to the UK or use of national security exceptions to block UK exports.	☑
Track record of high-quality IP protection.	In the US Chamber of Commerce's International IP Index 2023, Korea ranked 12th globally with an overall score of 84.44/100.[179]

Source: Adapted from Imagination Technologies and Global Counsel, 2022.[180]

5.2 Complementarities in semiconductors and AI

The table above demonstrates high-level suitability, but there is more to be gleaned at the specific level of semiconductors and AI. The fact that the UK and Korea have high ambitions in multiple technology areas means that there is an additional benefit to focusing partnership efforts on semiconductors and AI, technologies which are mutually reinforcing, and which provide benefits cascading through to other critical technologies.[181]

AI innovation, policy and regulation has been a site of shared activity and aspiration for both countries. Korea were important stakeholders in the UK's inaugural AI Safety Summit in November 2023, presenting their Digital Bill of Rights at the summit and expressing their support for transnational AI governance bodies. The next edition of the summit will be co-hosted by Korea and the UK in 2024. The UK's leading role in AI safety is regarded positively in Korea, providing an immediate platform for closer coordination and joint thought leadership.

[178] HM Government, *The Downing Street Accord: A United Kingdom-Republic of Korea Global Strategic Partnership* (Prime Minister's Office, 10 Downing Street: 22 November 2023), https://assets.publishing.service.gov.uk/media/655e58f602e2e1000d433691/November_2023_-_The_Downing_Street_Accord__A_United_Kingdom-Republic_of_Korea_Global_Strategic_Partnership.pdf.

[179] United States Chamber of Commerce, *International IP Index: 2023 Eleventh Edition* (Global Innovation Policy Center: 21 February 2023), 7, https://www.uschamber.com/assets/documents/GIPC_IPIndex2023_FullReport_final.pdf.

[180] Imagination Technologies and Global Counsel, *The Future of the UK's Semiconductor Strategy* (2022), 12.

[181] Ardi Janjeva, "A High Point in UK-South Korea Relations? Reflecting on President Yoon's State Visit and the Path Forward for Technology and Security Cooperation," *CETaS Expert Analysis* (December 2023).

On the semiconductors front, numerous interviewees recognised the natural complementarities between Korea's relative strengths in manufacturing (hardware) and the UKs in core IP and chip design (software).

> *'The UK has talent in the chip design space and Korea has capability in advanced manufacturing – there are interesting areas of collaboration there.'*[182]

> *'We've got the chip design capability and Korea have got the manufacturing. Already, ARM work closely with Samsung and SK Hynix, and Samsung is an investor in Graphcore.'*[183]

> *'Samsung have a historic presence in the UK with their design bases, the fact that they have facilities here already makes things easier.'*[184]

However, the software-hardware lens was seen by many as a starting point rather than the whole story. As per recommendations 5-10, there are six specific priorities for deepening **joint R&D and investment** between the two countries, three of which are discussed below:

- The UK has expressed a desire to develop **advanced packaging capabilities** to help build on strengths in compound semiconductors, photonics, and chip design.[185] Meanwhile, Korea is motivated to steer **new assembly, testing and packaging investment away from China.** While countries like Japan, Vietnam, Malaysia, and India are often referenced in this regard,[186] there is a distinct opportunity for **Korean investment into a 2.5D/3D hybrid advanced packaging centre in the UK**.[187]

- **UK research into novel forms of compute**, such as neuromorphic computing, quantum computing, and reversible computing is highly regarded.[188] None of these are ready to revolutionise the semiconductor industry today, but **the longer time horizon for development and commercialisation is in line with Korea's long-term investment planning.** For example, the $230bn investment in the Yongin chip cluster will not come online until 2042, by which point newer compute architectures may have proved their viability.

- **There are specific sub-domains of hardware and systems security research which the UK excels in, and Korea needs more of.** Mixed signal design capability is

[182] Author interview with government participants, 9 January 2024.
[183] Author interview with industry participant, 19 January 2024.
[184] Author interview with industry participant, 30 January 2024.
[185] HM Government, *National Semiconductor Strategy* (Department for Science, Innovation and Technology: May 2023), 36.
[186] Author interview with academic participant, 10 January 2024.
[187] Author interview with industry participant, 30 January 2024.
[188] Author interview with government participants, 9 January 2024.

not as prevalent in Asia, nor is the knowledge base around power and high voltage applications. Moreover, the UK's experience in radio frequency and advanced mixed signal technology would also likely be seen as desirable in Korea.[189]

5.3 Focusing efforts

Hitting the right challenges and achieving buy-in from UK-Korean industrial and academic bases is vital. One interviewee emphasised that researchers and government officials will be further away from the gaps and bottlenecks in the market than industry, and that devising bespoke mechanisms for exchange between academia and industry in both countries could yield practical solutions more quickly.[190] Lots of progress has been made at the government-to-government level via the Downing Street Accord, but this now has to be reflected in rates of FDI, dedicated funding for academic partnerships and joint ventures.

Nonetheless, both governments will still have a crucial role in setting the tone: "the market can be reluctant to take on early-stage risk, so government has to play that bridging role to kickstart the bigger, longer-term projects."[191] On the UK side, speeding up the development of semiconductor coordination capability through a new UK National Semiconductor Institute will be integral for broader strategic discussions among shared value chain partners.

Governments usually set different priorities for agreements with different partners. Korea's agreements with the US and EU respectively are an example of this: with the US there is a dedicated Export Control Working Group, whereas with the EU, members 'only exchange views on export controls' and there is greater emphasis on preparedness and crisis response.[192] Moreover, priorities can shift over time. The EU-US Trade and Technology Council initially focused on short-term supply chain issues, moving into subsidies and export controls, before more recently landing on collective research to address the environmental footprint of semiconductor manufacturing.[193] The lesson is that the structure of bilateral partnerships need tailoring and constant revision.

---

[189] Author interview with industry participant, 30 January 2024.
[190] Author interview with academic participant, 16 January 2024.
[191] Author interview with academic participant, 16 January 2024.
[192] Jan-Peter Kleinhans and John Lee, "Challenges of a Rising Chinese Chip Design Ecosystem," in *Europe's Strategic Technology Autonomy From China: Assessing Foundational and Emerging Technologies*, ed. Tim Ruhlig (Digital Power China: A European Research Consortium, January 2023), 27-40.
[193] Jan-Peter Kleinhans and John Lee, "Challenges of a Rising Chinese Chip Design Ecosystem," in *Europe's Strategic Technology Autonomy From China: Assessing Foundational and Emerging Technologies*, ed. Tim Ruhlig (Digital Power China: A European Research Consortium, January 2023), 27-40.

Both the UK and Korea share membership of various international organisations or alliances, and cooperating through these vehicles must complement bilateral discussions. One interviewee commented that, "the UK has a big influencing voice in the tech sector, and you do not leverage that just through bilateral engagements. You need to bring the bilateral engagement through with you into the bigger forums."[194]

Finally, numerous interviewees believed that the current golden period in UK-Korea relations provides an opportunity to advocate for each other's inclusion in new multilateral fora. For example, one interviewee commented that now the UK has joined the Comprehensive and Progressive agreement for Trans-Pacific Partnership (CPTPP), it would be a timely and strategic move to advocate entry into the Indo-Pacific Economic Framework led by the US.[195] On the other hand, the UK could put more energy into expanding the G7 to include countries like Korea and Australia: "Korea is at the centre of semiconductors, batteries, EVs, the whole range of emerging technologies. The G7 needs Korea's cooperation in these sectors."[196]

In summary, the UK and Korea must now establish a virtuous cycle between collaboration at the R&D stage and industry-based cooperation. Both Governments have made clear their intent for closer technology and security alignment, but targeted action is needed to ensure their academic and private sector communities buy in to the same vision and deliver on implementation at the ground level. If successful, the UK-Korea model of semiconductor-AI cooperation could become the blueprint for others to follow.

[194] Author interview with government participants, 9 January 2024.
[195] Author interview with academic participant, 16 January 2024.
[196] Author interview with academic participant, 16 January 2024.

Extended Recommendations

The table below builds on the Summary Recommendations by containing additional detail on the rationale behind each recommendation and category of partnership.

Category of partnership	Rationale	Recommendation
Establishing and coordinating national authorities	Although the UK and Korean governments have been pursuing closer strategic cooperation at the leadership level, there is work to be done to ensure this is prioritised across the whole government stakeholder ecosystem.	1. **A UK National Semiconductor Institute should be established this year.** Working with the UK Semiconductor Advisory Panel, this body should develop a bespoke engagement strategy with Korea's Advanced Semiconductor Technology Centre.
Coordinating non-government expertise	UK-Korea industry and academic relationships need more proactive support to meet the high bar set in recent agreements at the government level. Given geographical disparities, time zone differences, and more limited day-to-day cultural engagement, these relationships will need nurturing to flourish in the long-term.	2. DSIT and MSIT should sponsor **quarterly meetings between semiconductor industry stakeholders in both countries, expanding this to academic and research institutes as necessary**. The next step could then be for DSIT and MSIT to develop a joint fund for UK/Korean AI chip start-ups.
Coordinating AI and semiconductor portfolios	The new '2+2' structure bringing together the Foreign and Defence Ministries of the UK and Korea annually could be mirrored for the	3. The UK and Korean Governments should **develop joint working groups for departments overseeing AI and semiconductor policy respectively (DSIT and the**

	departments chiefly responsible for AI and semiconductors. Formalising these engagements would allow for easier sharing of best practice regarding initiatives like MSIT's AI Semiconductor Scaling Network.	**Department for Business and Trade [DBT] in the UK, and MSIT and the Ministry of Trade, Industry and Energy [MOTIE] in Korea).**
AI-enabled early warning systems	The complexity and opacity of semiconductor supply chains makes it difficult for individual countries to identify and forecast the whole range of supply chain risks. The development of AI will allow for a greater expansion of supply chain forecasting and anticipation of bottlenecks.	4. Korea's MOFA (via the government's new Supply Chain Stability Committee) and the UK's DBT should **develop more formal, real-time information sharing mechanisms relating to supply chain risks and vulnerabilities**, cross-referencing with the OECD's Semiconductor Informal Exchange Network.
Government funding	Korean government departments such as MSIT are now making more money available for international research collaborations, marking a shift from the status quo where such funding was predominantly restricted to Korean companies and institutes.	5. **DSIT and MSIT should ringfence proportions of their international funding for collaborations between UK and Korean research institutes or companies**. Initial funding should prioritise longer-term projects to target the development of mid-career researcher relationships. This initiative could also involve a 'Technology Transfer and Commercialisation' scheme between the two countries.
Collaboration on new AI compute architectures	Both the UK and Korea have high-potential companies which are	6. The UK and Korean governments, working closely with industry, should **promote**

	focusing exclusively on developing new AI compute architectures (Graphcore IPUs and Sapeon NPUs) and could grow faster by pooling relevant expertise and exchanging best practice.	**strategic partnerships between companies best positioned to develop GPU-alternative future compute infrastructures**. This could be modelled on the Samsung Austin Research Centre (SARC) in Texas.
Advanced packaging facilities	The UK has stated an intention to build a limited amount of advanced packaging capability from scratch. Korean industry should view this as an opportunity to support their friend-shoring objectives and avoid an over-reliance on China for a stage of the supply chain that is vulnerable to malicious interventions.	7. **A UK National Packaging Institute** should be established, which could be bid for by existing industrial, governmental and university organisations in the UK. The new Institute should **explore the suitability of Korean investment into a new 2.5D/3D hybrid advanced packaging centre in the UK,** while partnering with Korean companies on wafer scale integration. This could be modelled on Samsung's recent investment in a new advanced packaging centre in Yokohama, Japan.
Reserve capacity	The UK is dependent on industrial policy efforts amongst its allies succeeding so that it can mitigate the most severe effects of a China/Taiwan crisis. Tasking industry partners with specific objectives and targets is one way of ensuring the UK gains from indirect investment into its own supply via partners in Korea.	8. UK and Korean national semiconductor bodies should liaise with industry to determine whether a **percentage of Korean fabrication capacity can be dedicated to servicing UK start-ups**; in return UK industry should scope which areas of its design capacity could be marketed to Korean companies on a short timescale. This agreement could target manufacturing

		capacity that has not yet come online, such as the proposed mega-cluster near Seoul.
Compound semiconductors	The UK has recognised strengths in compound semiconductor R&D but there is a gap in the subsequent commercialisation. There is a need to bridge the journey between developing compound semiconductor devices and volume manufacture. At the same time, Korean investment in the UK market could be a way of offsetting a future Chinese monopoly.	9. The UK should pursue plans to create an **open-access compound semiconductor foundry** and **assess the feasibility of Korean investors making up any funding shortfall**. The UK compound semiconductor industry should leverage its leading research base to attract investment as Korean industry leaders seek to build out their compound semiconductor lines.
Cybersecurity/hardware expertise	There are specific sub-domains of hardware research which the UK excels in, and South Korea needs more of. This expertise is spread across UK academia and the defence and security community. Greater cooperation in this domain represents an important opportunity in that it is not tied to specific manufacturing pathways adopted by either country.	10. As part of the new UK-Korea Strategic Cyber Partnership, the UK Ministry of Defence and Korean defence and security community should **scope hardware security or capability domains that could be suitable for joint R&D or sharing of best practice**, such as radio frequency and advanced mixed signal technology. This could also be mirrored in a cross-country industry/academia partnership which invests in technical solutions to screen malicious hardware.
Design IP/tooling capability	The high costs of EDA tools are a significant barrier to entry in the UK	11. The UK Government should fund **a new Design Competence Centre, with**

	semiconductor market and impacts on development and production timelines for new chip designs.	**Korea as a priority partner upon international scaling.** This could launch a **UK-Korea moonshot project to create AI-based open-source EDA tools** for all aspects of chip design.
Skills	Skills shortages continue to be endemic to the semiconductor industry. Different countries will have bottlenecks in different areas of the supply chain. Having an agreement for easier exchange of skilled personnel between the two countries in this area would be transformational.	12. UK and Korean industry leaders should conduct a **joint mapping exercise identifying skills shortages in their respective semiconductor sectors and devise a two-way upskilling programme** which pools resources into areas with the greatest need. This could evolve into a public-private partnership aimed at developing 'lab-to-fab' talent linkage.
Third countries and multilateral fora	Translating positive agreements at the bilateral level into continued support at the multilateral level will help drive global consensus in the direction of UK-Korea priorities. This should leverage the new annual 2+2 Foreign and Defence Ministerial meetings announced in the Downing Street Accord.	13. The UK and Korea should strive for **closer policy alignment in third countries with a strategic position in the semiconductor supply chain,** and each consider how they can **advocate for the expansion of certain multilateral fora or agreements to include the other.**
International standards	The UK and Korea have a strong interest in activities at international standards bodies, and there are various opportunities to leverage the cybersecurity	14. **UK-Korea cybersecurity cooperation should be reflected in their positioning on international standards.** The existing R&D partnership between DSIT in the UK and

	strengths in the UK and telecoms strengths in Korea to achieve mutually beneficial results in these arenas.	the IITP in Korea designed to accelerate the deployment of Open RAN technologies should be scaled up, taking advantage of Korean industry's central positioning.
AI safety	Both the UK and Korea are seeking to lead global discourse and action on AI safety. As well as cooperation on principles, this will require innovative solutions to ensure that both governments have the requisite compute.	15. **When Korea launches its AI Safety Research Institute, the UK's AISI should be its leading international partner**, aligning closely with the US AISI based in NIST and partners in Europe to **share best practice and develop common approaches.**

About the Authors

Ardi Janjeva is a CETaS Research Associate at The Alan Turing Institute. His research interests include technology-enabled threats in the 21st century; the future of intelligence innovation; technology-based geostrategic alliances and competition; and the relationship between technology and economic resilience.

Professor Seungjoo Lee is a CETaS Visiting Fellow and Professor of Political Science and International Relations at Chung-Ang University (Seoul, Korea). His current research focuses on the economy-security nexus, economic statecraft, US-China technology competition, and global digital governance. He currently serves as a member of the Advisory Committee on Economic Security and Foreign Affairs (Ministry of Foreign Affairs, Korea) and a member of the Policy Advisory Board of the Ministry of National Defense.

Professor Harish Bhaskaran is a Professor of Applied Nanomaterials at the University of Oxford, where he directs the Oxford Fab, and serves as the Associate Head for Research of the Mathematical, Physical and Life Sciences Division. He is distinguished for his research achievements in engineering nanoscale devices including breakthrough photonic or neuromorphic computing and displays.

Dr Seoin Baek is an Assistant Professor of the Department of Chinese Studies/Department of Robotics at the Hanyang University ERICA campus. Before joining Hanyang University, he worked as Associate Research Fellow/Head of Science & Technology Diplomacy Policy Research Office at the Science and Technology Policy Institute (STEPI). His research investigates innovation policy and strategy at firm/industry/country level.

Hyunjin Lee is a Research Fellow at the Overseas Economic Research Institute of the Export-Import Bank of Korea (KEXIM). His current responsibilities encompass industry analysis and advisory on artificial intelligence and 5G technology. He has served as an external expert contributor and advisor to major organisations, including the Ministry of Foreign Affairs and other national research institutes. His recent research utilises data-driven methodologies to address geopolitical effects on high-tech industries.



Centre for Emerging Technology and Security

RESEARCH REPORT